

# Linear Models

## Geometric Algorithms

# Practice Problem

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$

*Find the projection of  $\mathbf{b}$  onto  $\text{Col}(A)$*

*(note that  $\mathbf{a}_3 = \mathbf{a}_2 - \mathbf{a}_1$ )*

**Answer**

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$



# Objectives

1. Use the least square method to build linear *models* of noisy data
2. Show how we can use linear algebraic methods to model with non-linear models

# Keywords

line of best fit

independent/dependent variables

residuals

prediction

simple least squares regression

multiple regression

polynomial regression

models

model fitting

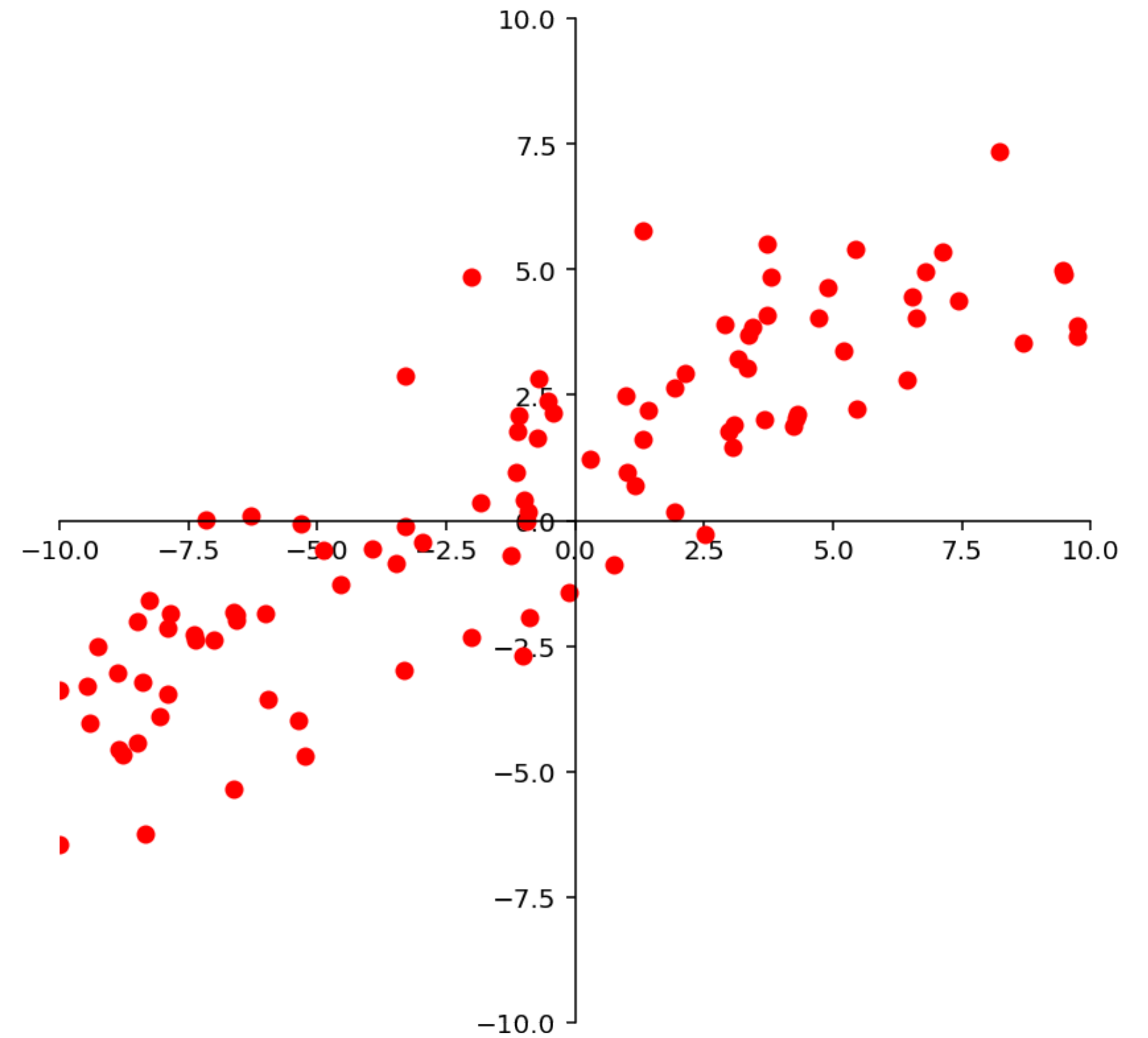
model parameters

design matrices

# **Warm-up: Line of Best Fit**



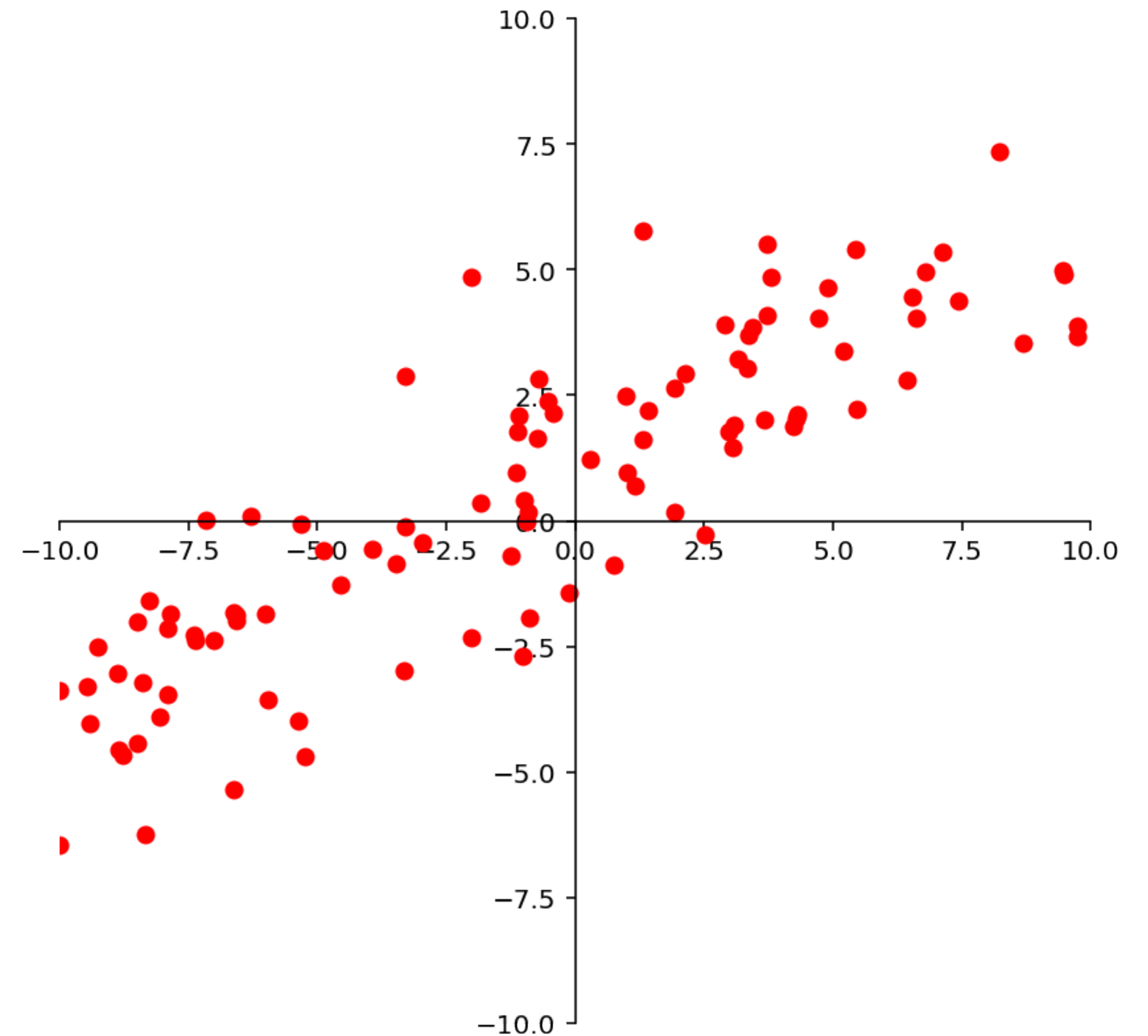
# The Setup



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You're given points in  $\mathbb{R}^2$ :

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$



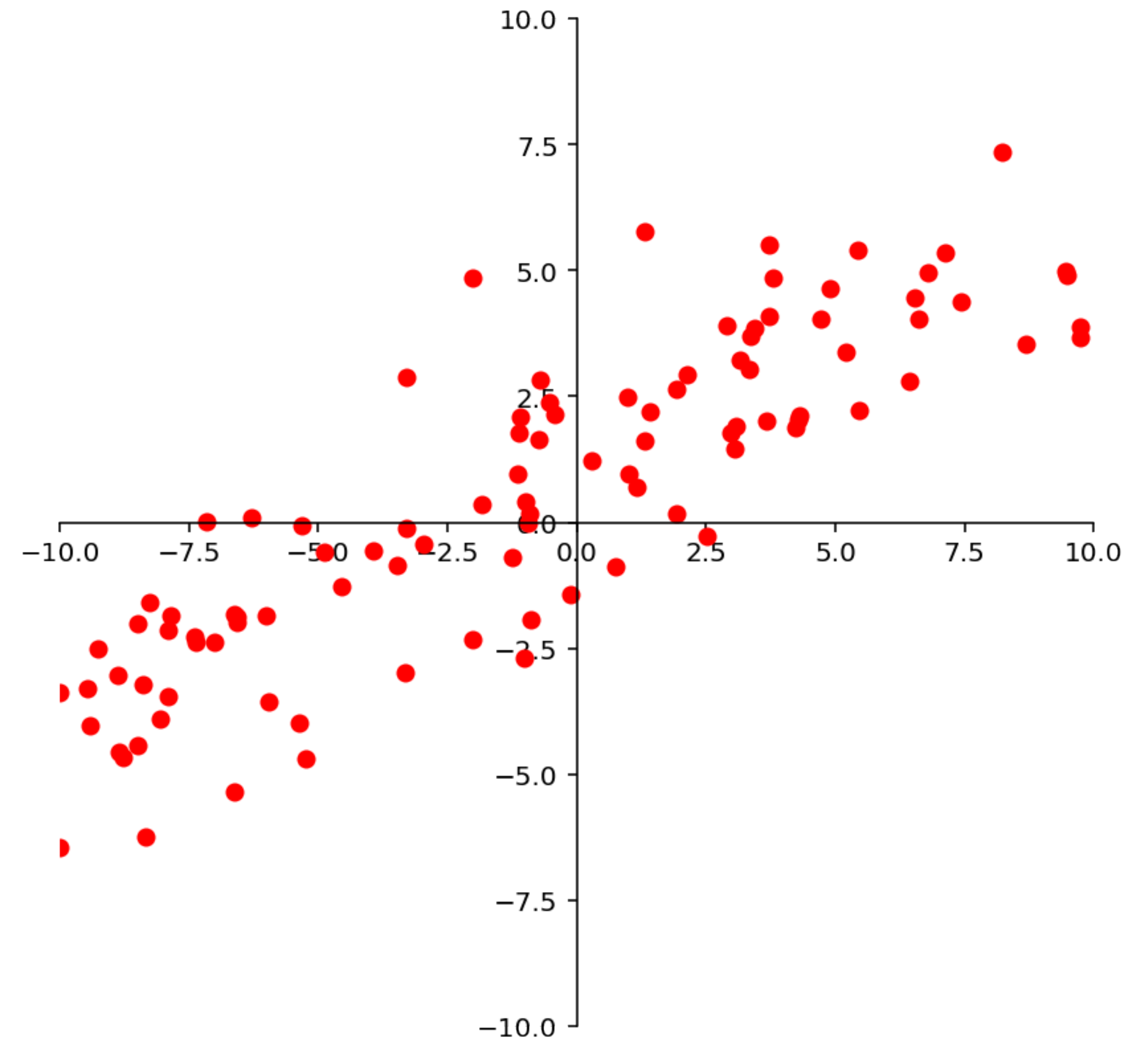


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**Ex.** (height, weight) data for a population



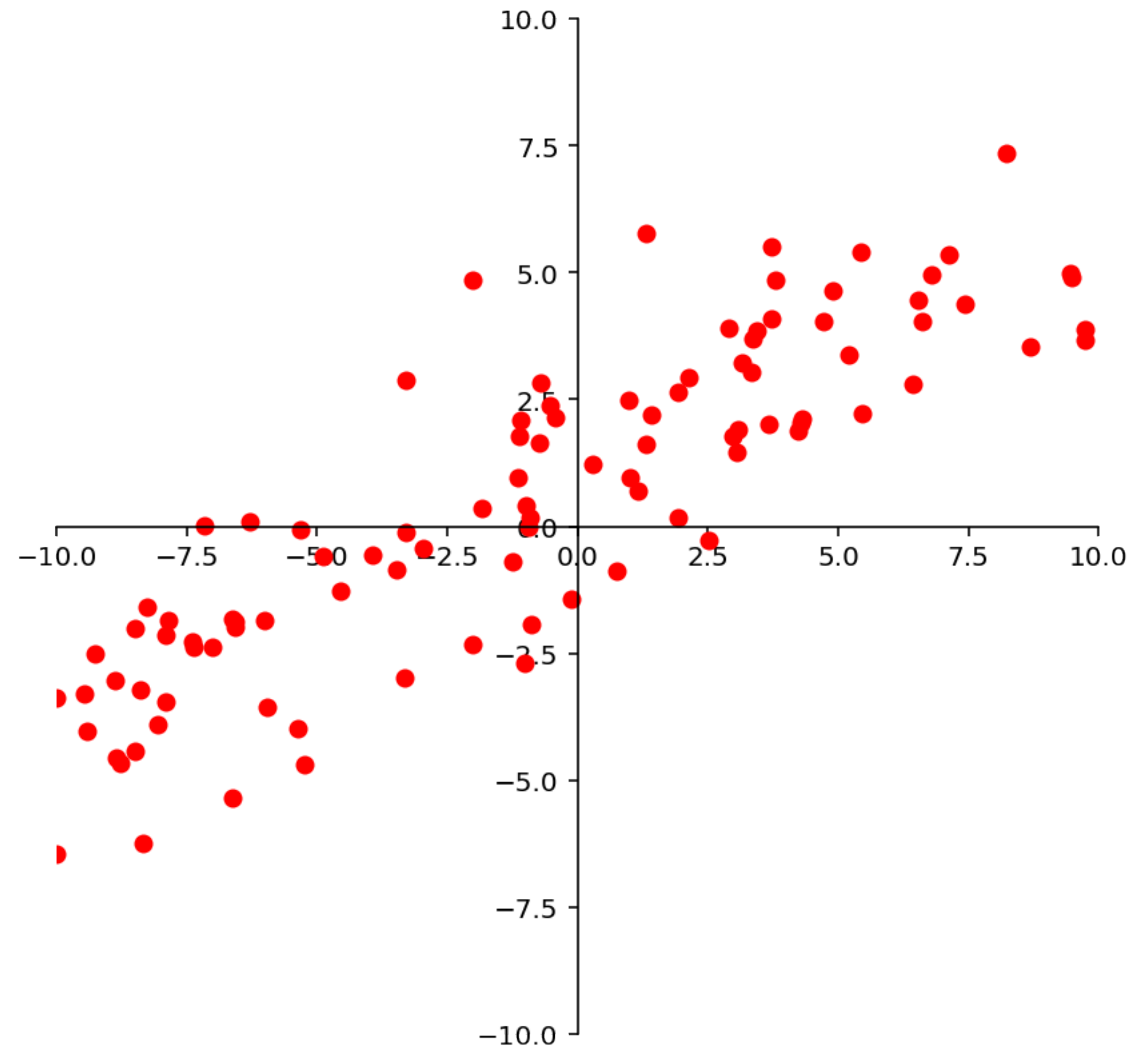
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The data *kind of* trends  
as a line



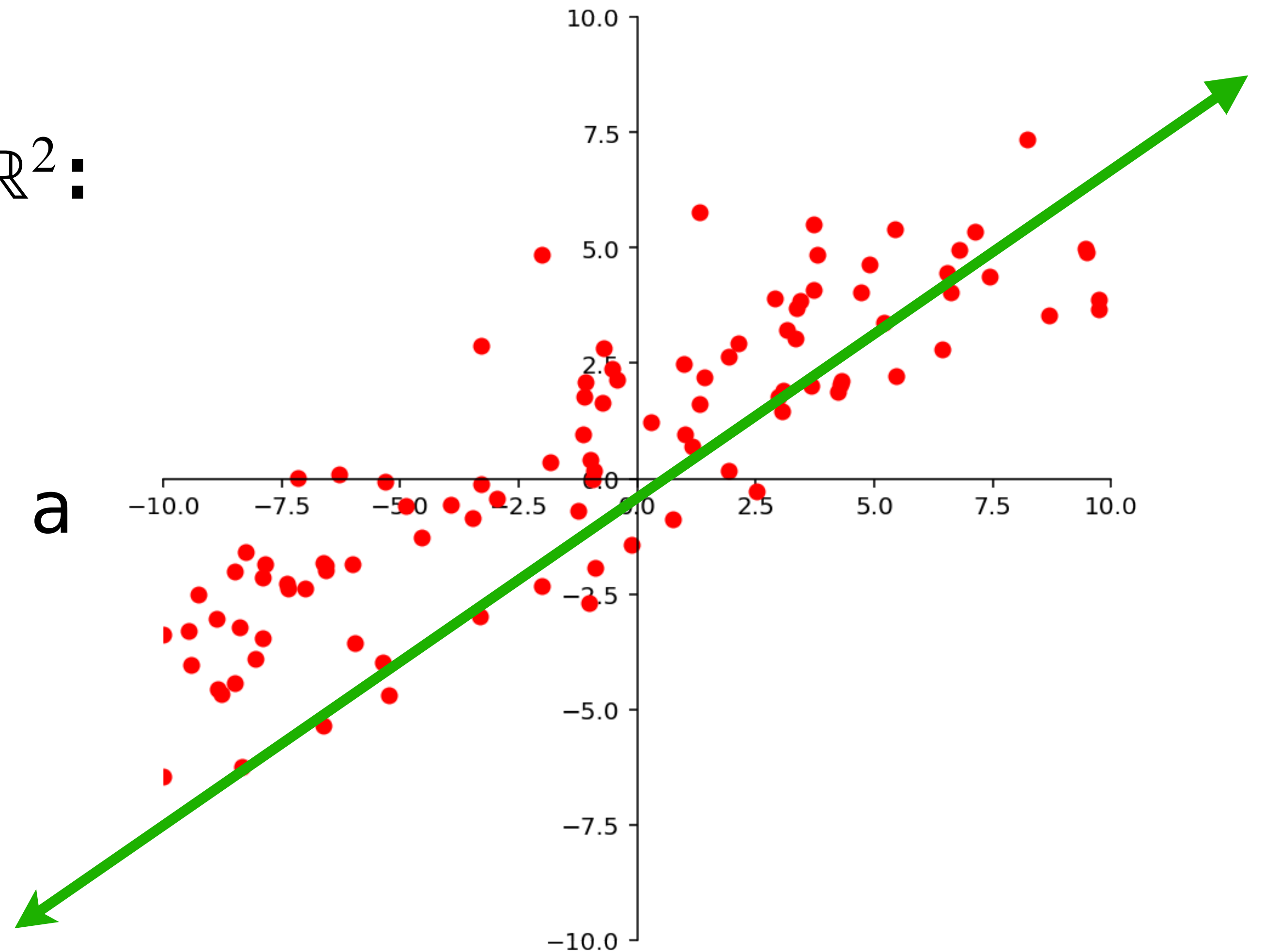
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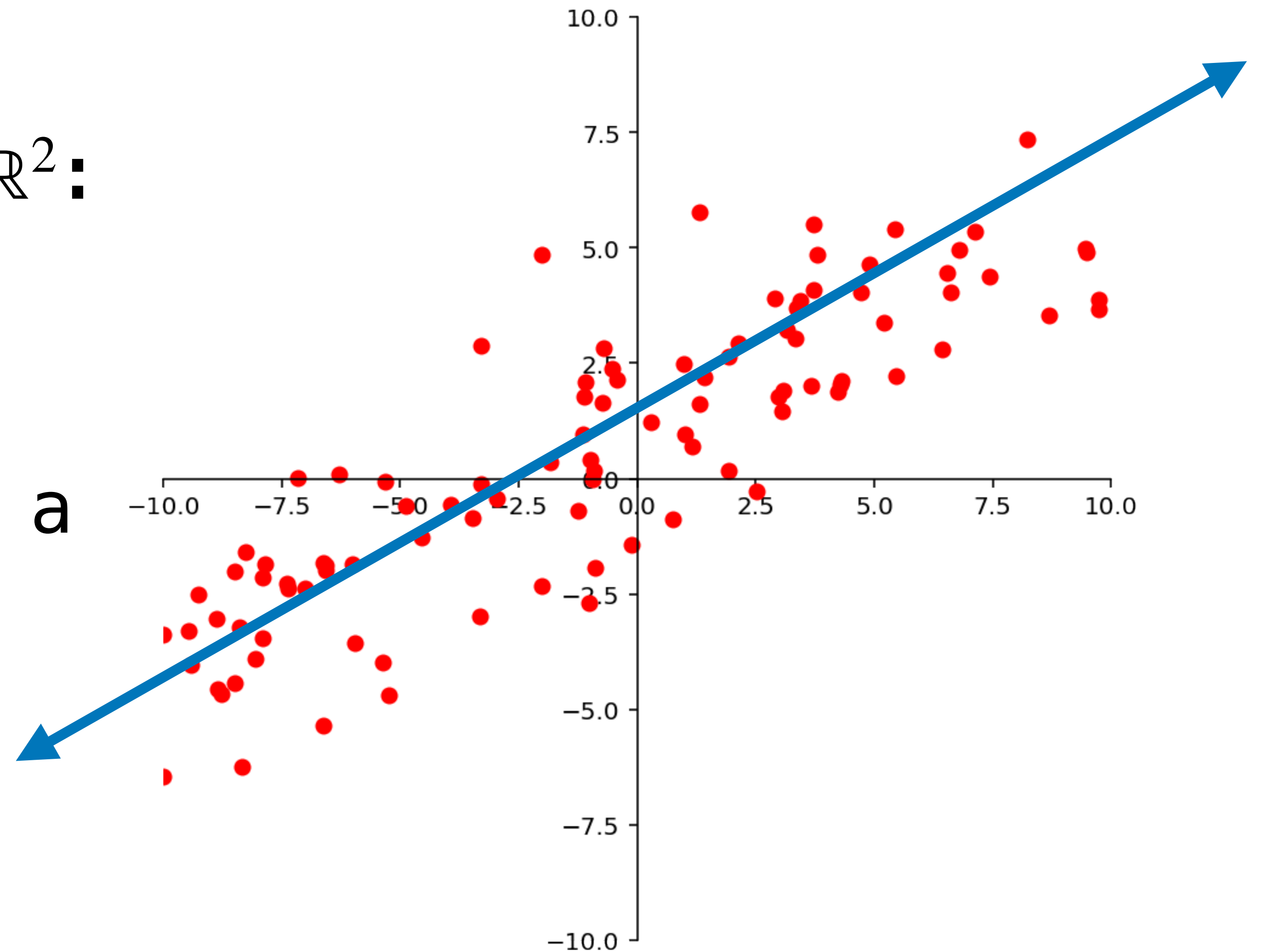
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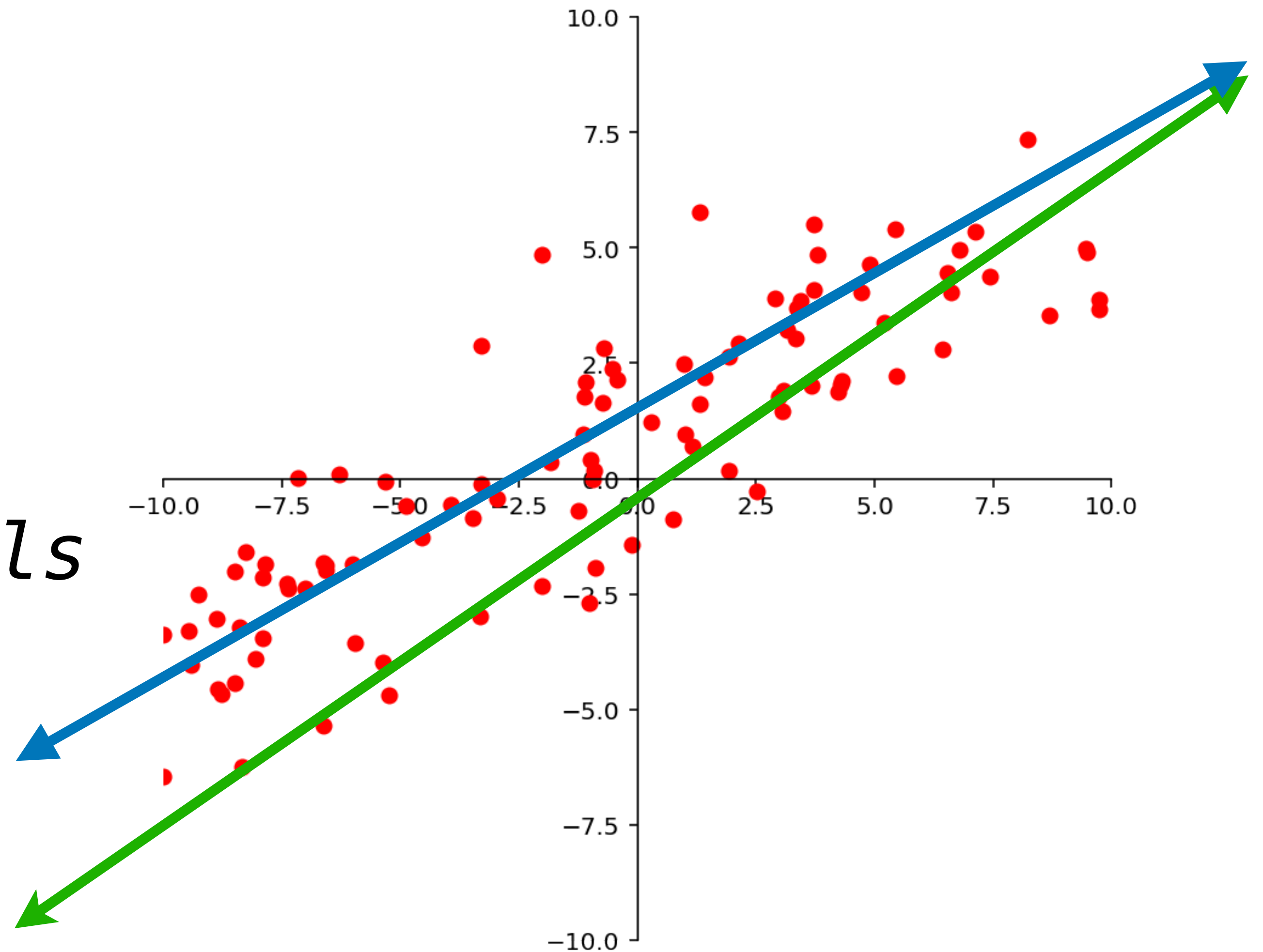
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# The Setup

Q. Which line "best" describes the trend of the dataset?

Q. Which one best models the dataset?



# Two Important Questions



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**We'll come back to this...**



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1. What is a model?

**We'll come back to this...**

2. What does "best" mean?

**This is a make-or-break question**

# Least Squares Simple Linear Regression

**Q.** Given a set of points  $\{(x_1, y_1), \dots, (x_n, y_n)\}$ , find the line

$$f(x) = \beta_0 + \beta_1 x$$

which *minimizes*

$$\sum_{i=1}^n (y_i - f(x_i))^2$$



# Least Squares Simple Linear Regression

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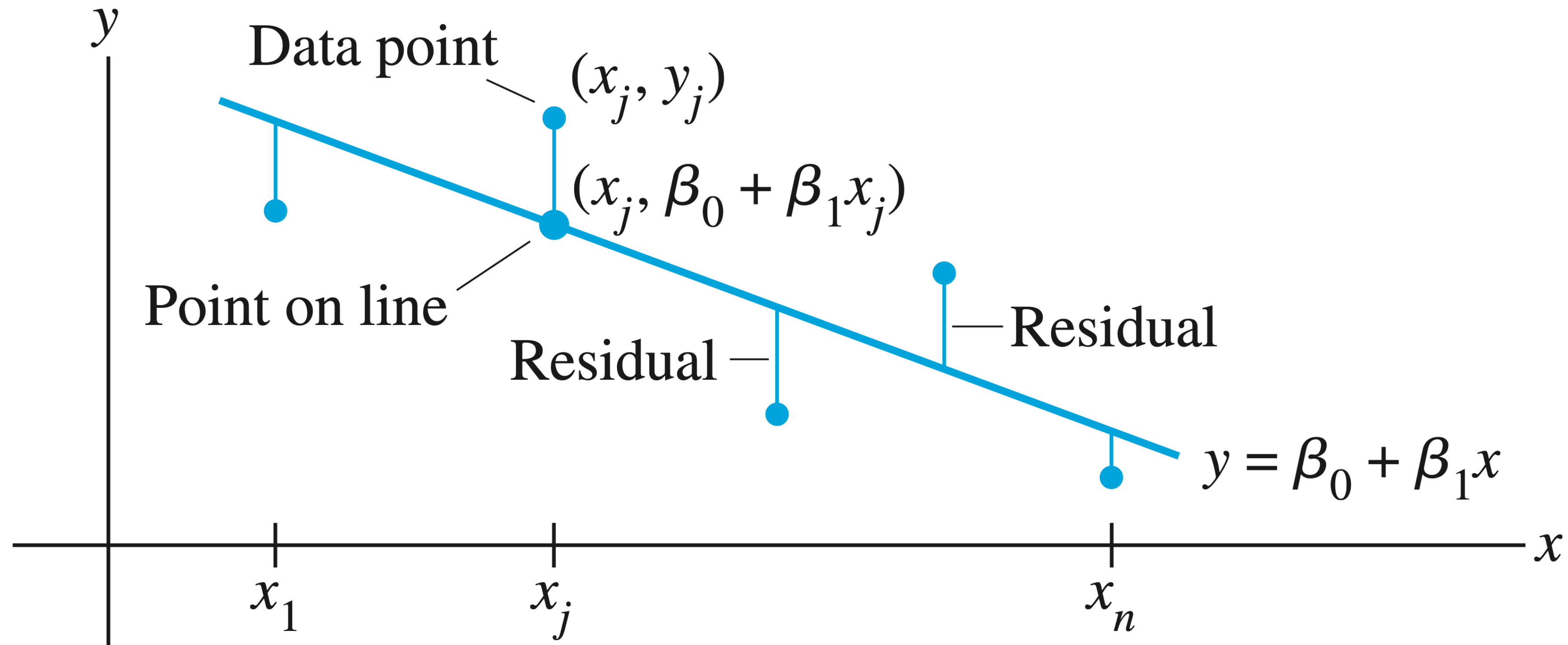
$$f(x) = \beta_0 + \beta_1 x$$

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The "best" line  
minimizes the *sum  
of squares of  
differences*

# The Picture



Find the line which makes the (squared) sum of these differences *as small as possible*

# An Aside: Is this really "best"?





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*Who's to say...*



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Depends on the data, the  
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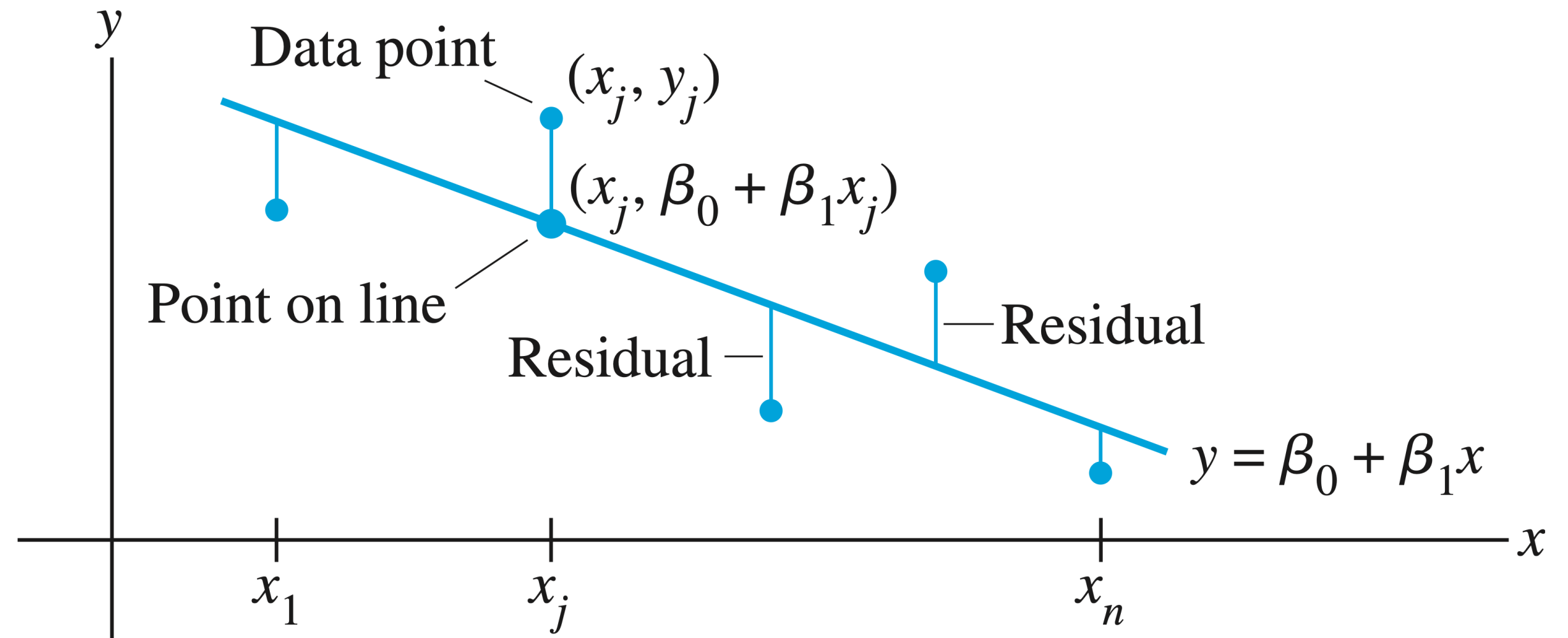
*Who's to say...*

Depends on the data, the application domain, etc.

**The point.** We fix the notion of "best" and then we do calculations



# Terminology



$$\{(x_1, y_1), \dots, \text{data point } (x_i, y_i), \dots, (x_n, y_n)\} \text{ dataset}$$

$$f(x) = \text{model parameters / regression coefficients } \beta_0 + \beta_1 x \text{ model}$$

independent variable  $x_i$   $y_i$  dependent variable (label)

A dashed arrow points from the data point  $(x_i, y_i)$  in the dataset to the variables  $x_i$  and  $y_i$  in the regression equation.

$$\sum_{i=1}^n \text{observation } y_i - \text{prediction } f(x_i) \text{ residual } ^2$$

# Terminology: Datasets

$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

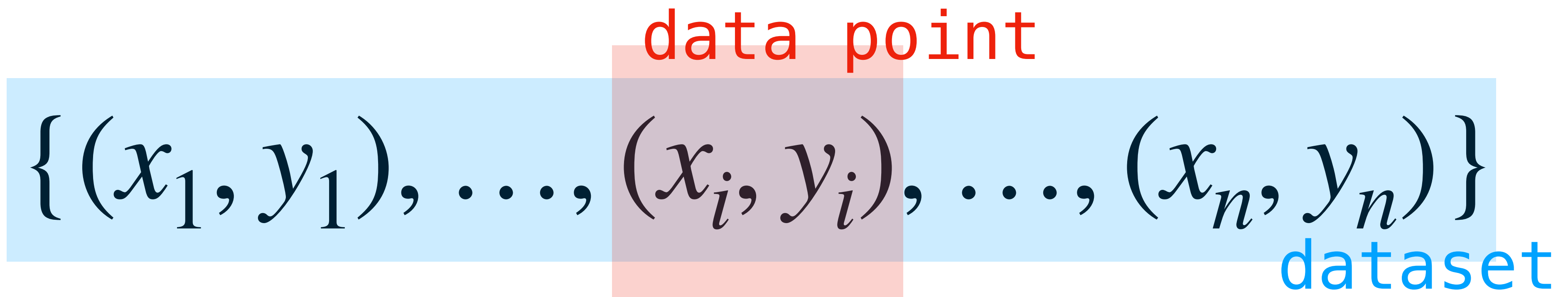


# Terminology: Datasets

$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

dataset

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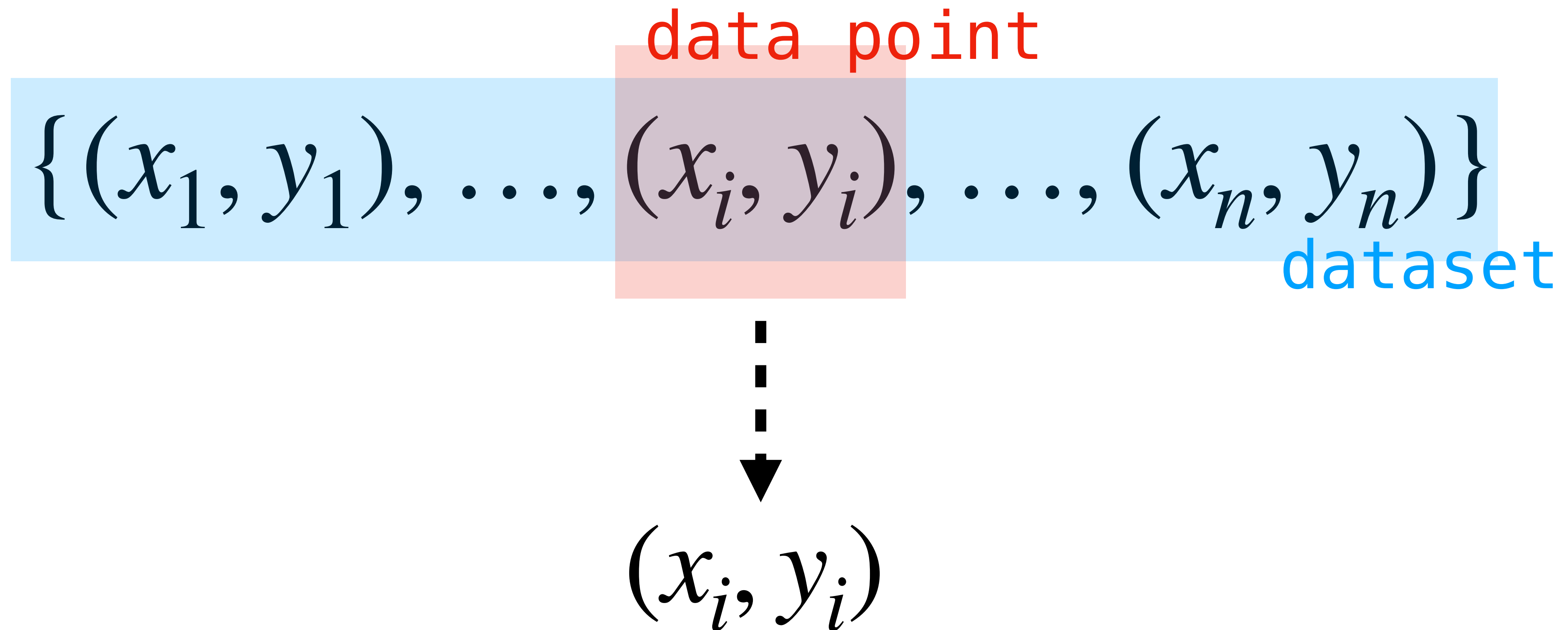
The diagram illustrates the terminology for datasets. It features a light blue rectangular background representing the entire dataset. Inside this rectangle, the mathematical expression  $\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$  is written in a black serif font. A smaller, semi-transparent light red rectangle is positioned over the term  $(x_i, y_i)$  within the set notation. Above this red rectangle, the text "data point" is written in red. Below the right side of the blue rectangle, the word "dataset" is written in blue.

data point

$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

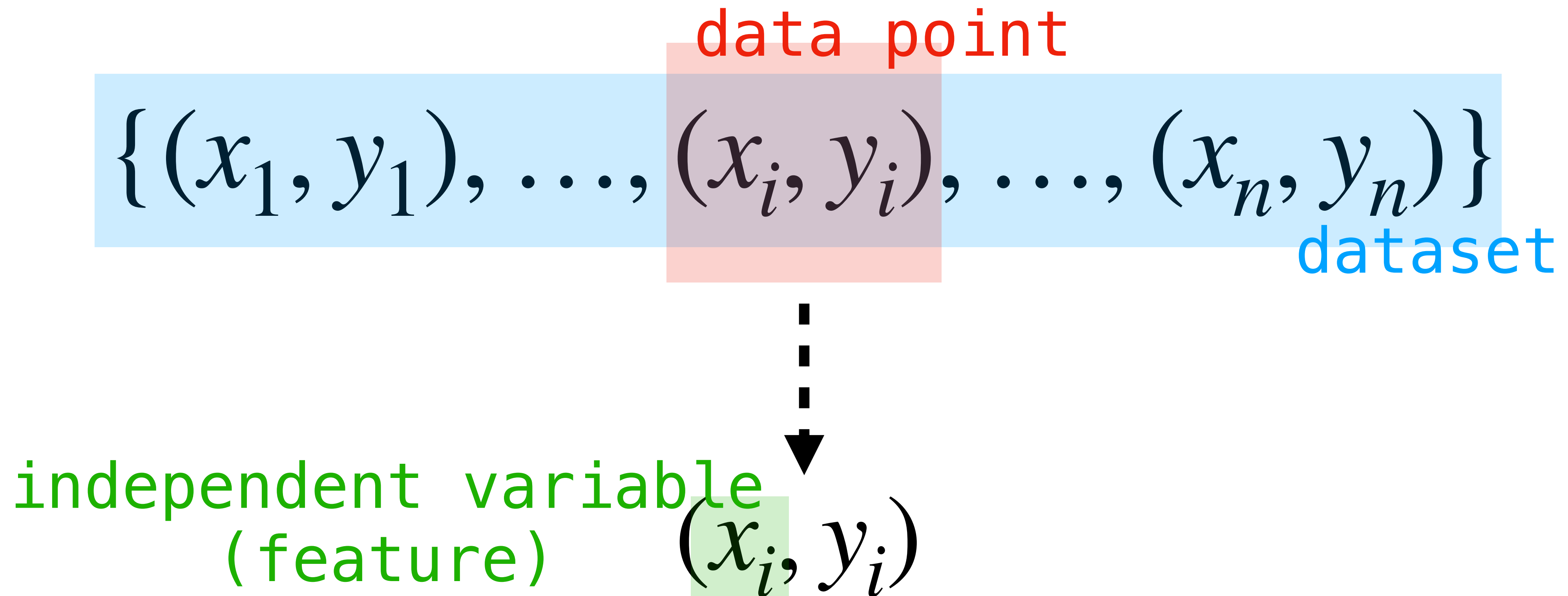
dataset

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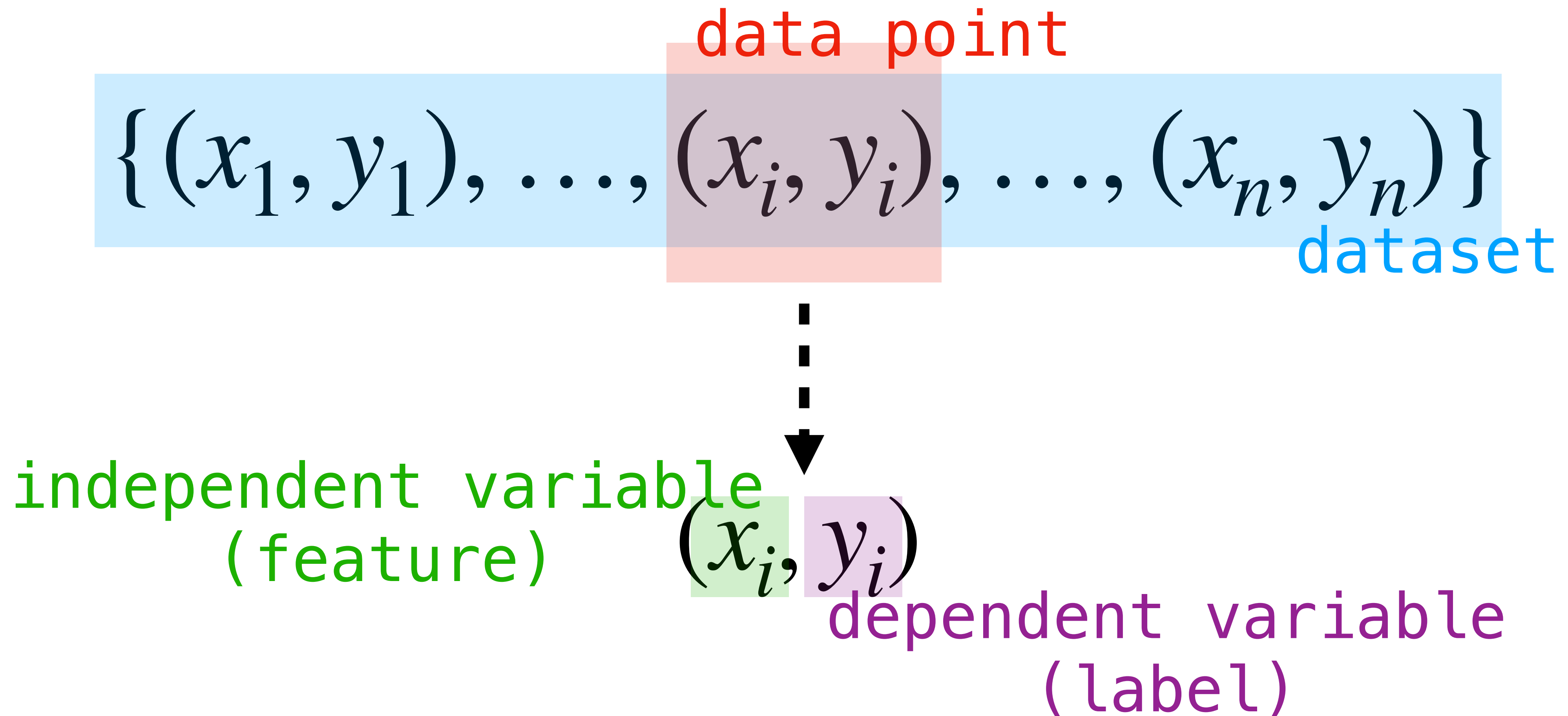




# Terminology: Datasets



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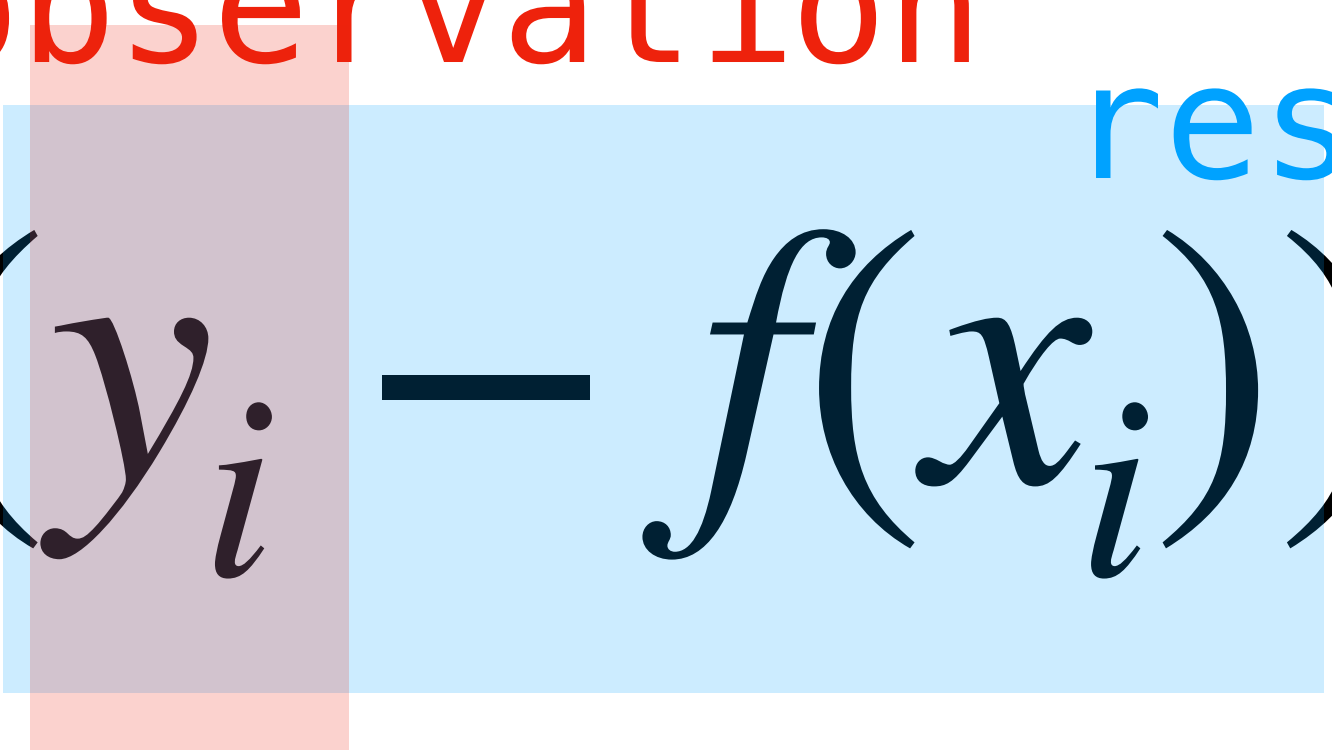
residual

# Terminology: Least-Squares Error

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

observation

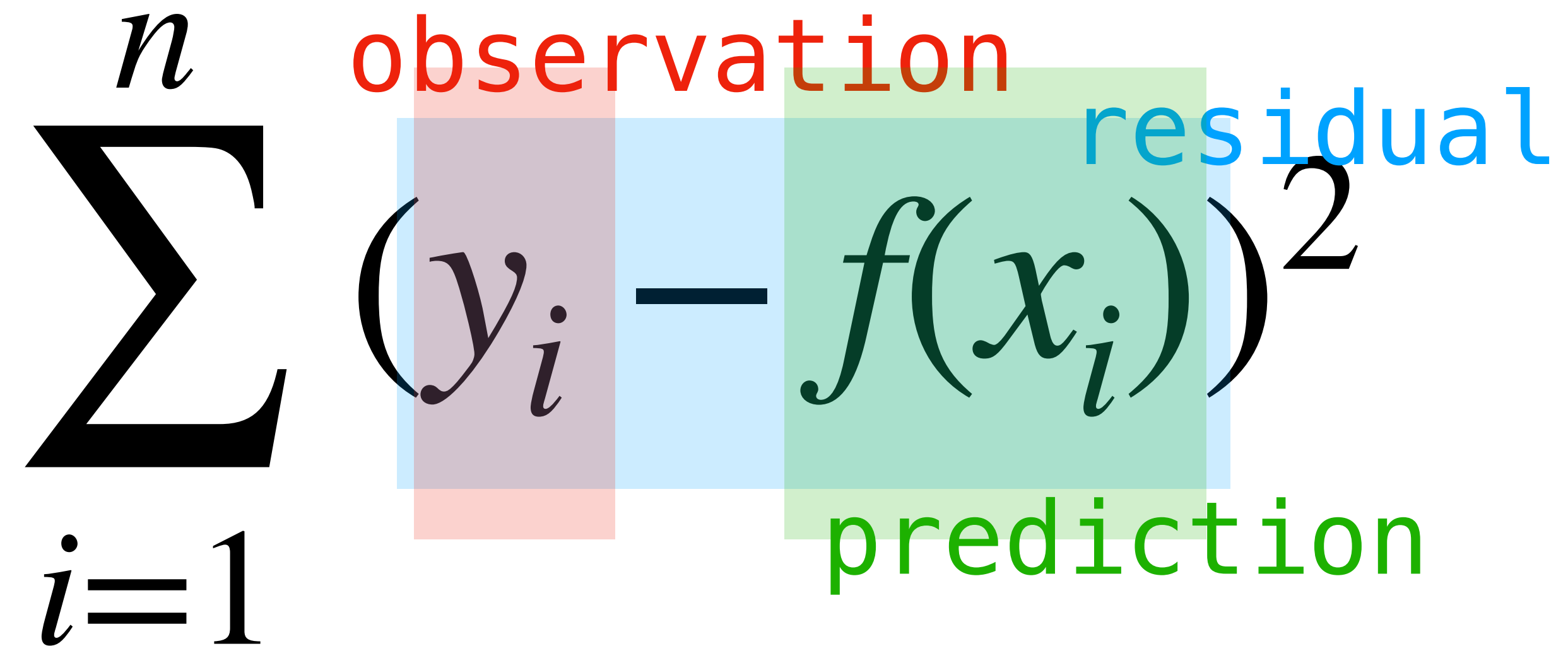
residual



The diagram illustrates the components of the least-squares error formula. The summation symbol  $\sum$  is positioned to the left of the expression  $(y_i - f(x_i))^2$ . The index  $i$  is below the summation symbol, and the upper limit  $n$  is above it. The term  $y_i$  is enclosed in a light red rectangular box, with the word "observation" written in red above it. The term  $f(x_i)$  is enclosed in a light blue rectangular box, with the word "residual" written in blue above it. The minus sign and the exponent  $^2$  are not enclosed in any boxes.



# Terminology: Least-Squares Error



The diagram illustrates the least-squares error formula with color-coded components. The summation symbol  $\sum$  is on the left, with the upper limit  $n$  and the lower limit  $i=1$ . The formula is  $(y_i - f(x_i))^2$ . The term  $y_i$  is enclosed in a light red box labeled "observation" in red text above it. The term  $f(x_i)$  is enclosed in a light green box labeled "prediction" in green text below it. The entire expression  $(y_i - f(x_i))^2$  is enclosed in a light blue box labeled "residual" in blue text above it.

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

# How to: Finding the Least Squares Line

$$\beta_1 = \frac{n \sum_{i=1}^n x_i y_i - \left( \sum_{i=1}^n x_i \right) \left( \sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left( \sum_{i=1}^n x_i \right)^2} \quad \beta_0 = \frac{\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i}{n}$$



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**A (First attempt).** Use these equations...

# How to: Finding the Least Squares Line

Don't memorize these

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# An Observation

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^n ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

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minimize for least-squares line

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*These expressions look very similar*

**Can we *design* a matrix where finding a least squares solution gives us a least squares line?**

# A Least Squares Problem

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$$\vdots$$

$$\beta_0 + \beta_1 x_n = y_n$$



# A Least Squares Problem

In the "ideal" world, we could find  $\beta_0$  and  $\beta_1$  such that all of these equations hold

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This would mean **all the points already lie on a single line**

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$$\begin{aligned}\beta_0 + \beta_1 x_1 &= y_1 \\ \beta_0 + \beta_1 x_2 &= y_2 \\ &\vdots \\ \beta_0 + \beta_1 x_n &= y_n\end{aligned}$$

**This is a linear system in the variables  $\beta_0$  and  $\beta_1$**



# A Least Squares Problem

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

# A Least Squares Problem

In the "ideal" world,  
*this matrix equation*  
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In the "ideal" world,  
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In reality it won't  
have a solution, **but we**  
**can find an approximate**  
**solution**

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# A Least Squares Problem

$$\begin{bmatrix} 1 & \overset{\textcolor{red}{X}}{x_1} \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \overset{\textcolor{red}{\vec{\beta}}}{\beta_0} \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \overset{\textcolor{red}{y}}{y_1} \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\|X\vec{\beta} - \mathbf{y}\|^2 = \sum_{i=1}^n ((\beta_0 + \beta_1 x_i) - y_i)^2$$

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The sum of squares of residuals is the squared distances between  $X\beta$  and  $\mathbf{y}$



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*Least squares solutions to this system give us parameters for least squares lines*

# Recall: The Normal Equations

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**Thm.** The set of least-squares solutions of  $A\mathbf{x} = \mathbf{b}$  is the same as the set of solutions to

$$A^T A \mathbf{x} = A^T \mathbf{b}$$



# Recall: The Normal Equations

**Thm.** The set of least-squares solutions of  $A\mathbf{x} = \mathbf{b}$  is the same as the set of solutions to

$$A^T A\mathbf{x} = A^T \mathbf{b}$$

**In particular, this set of solutions is nonempty**

# Recall: Unique Least Squares Solutions

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

If  $A$  has LI columns, then its unique least squares solution is defined as above

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

# How To: Least Squares Line

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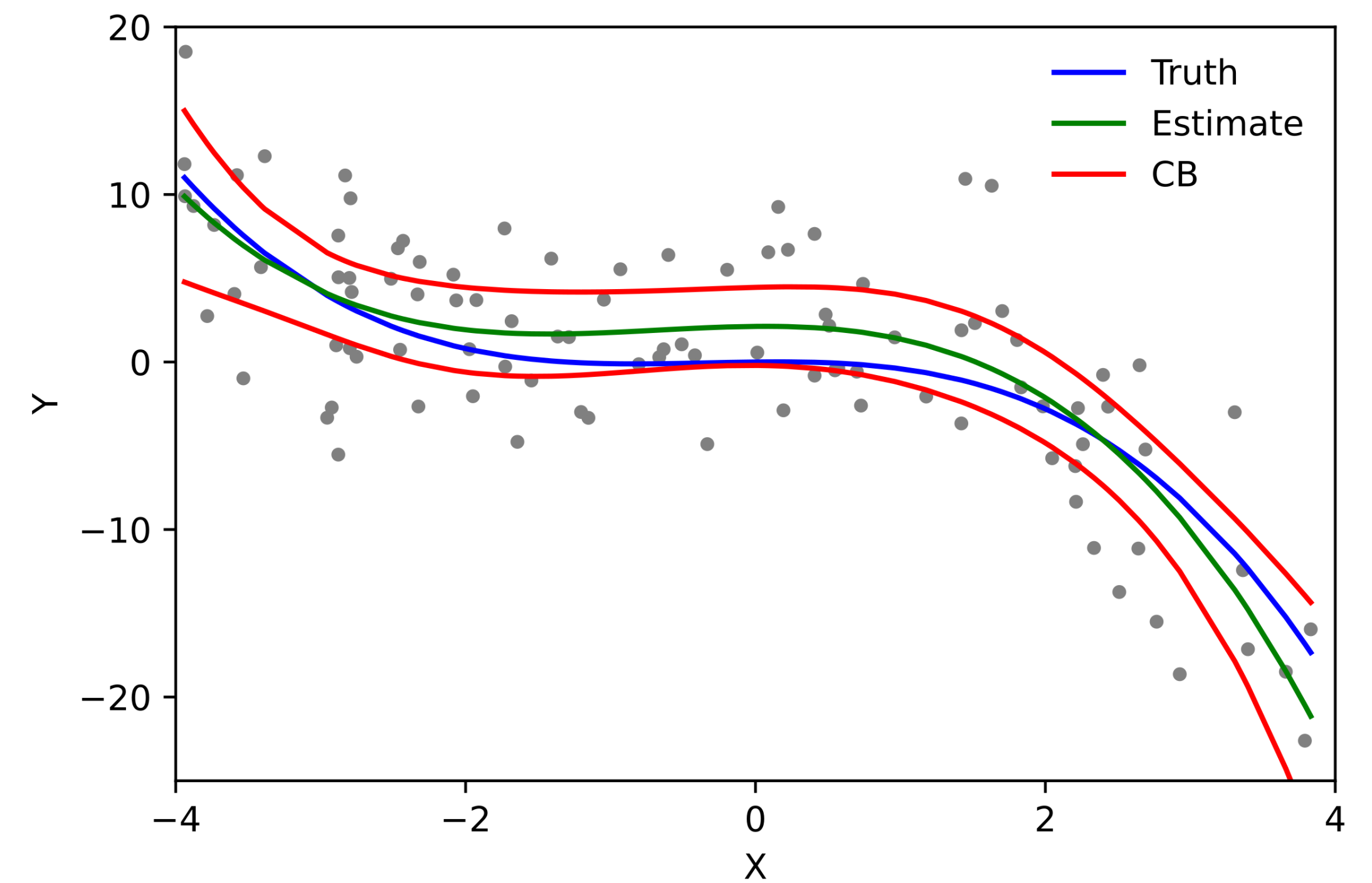
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**Q.** Find the least squares line for the dataset  $\{(x_1, y_1), \dots, (x_n, y_n)\}$

**A.** Find the least squares solution to the above equation

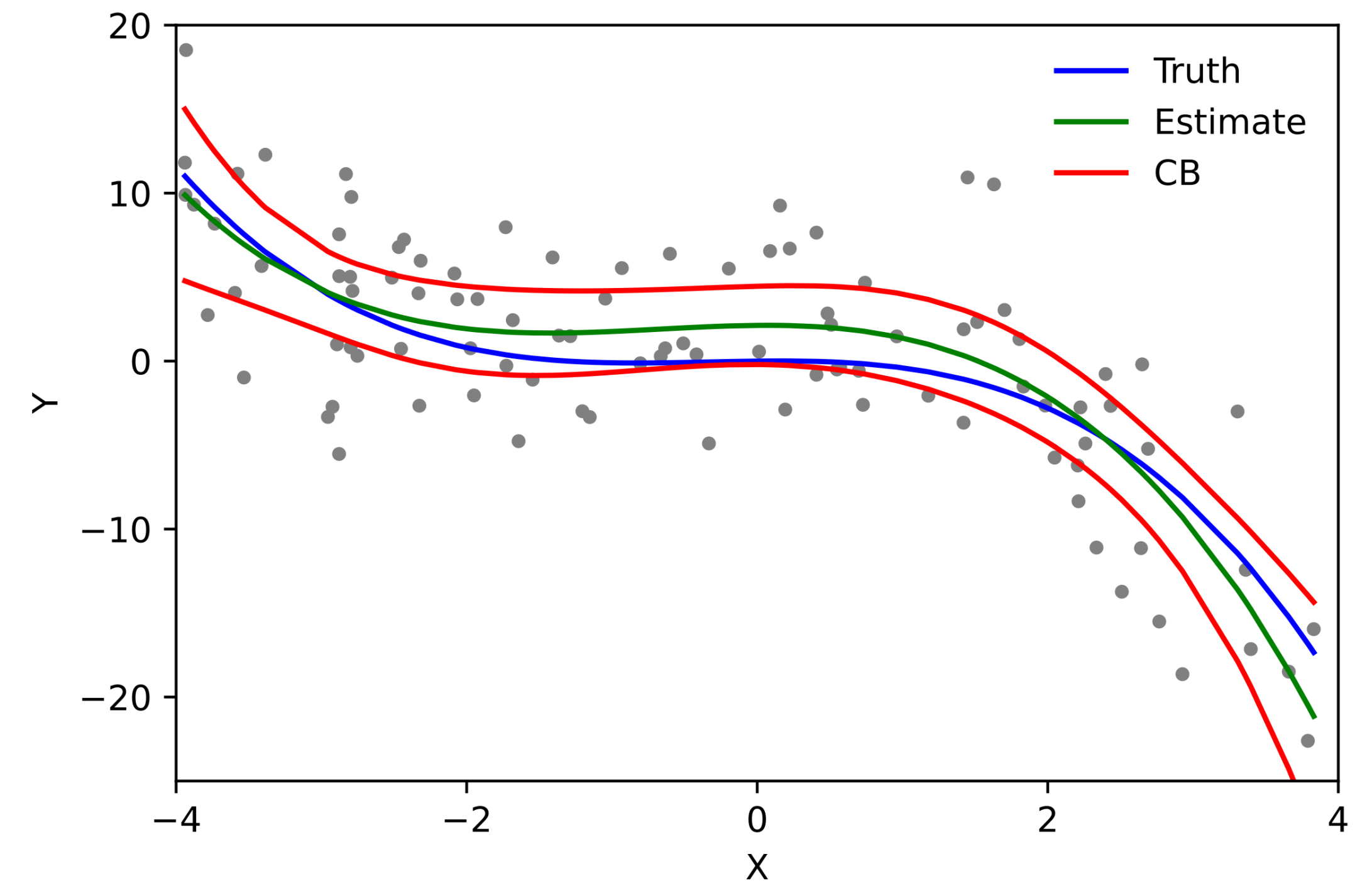


# General Regression



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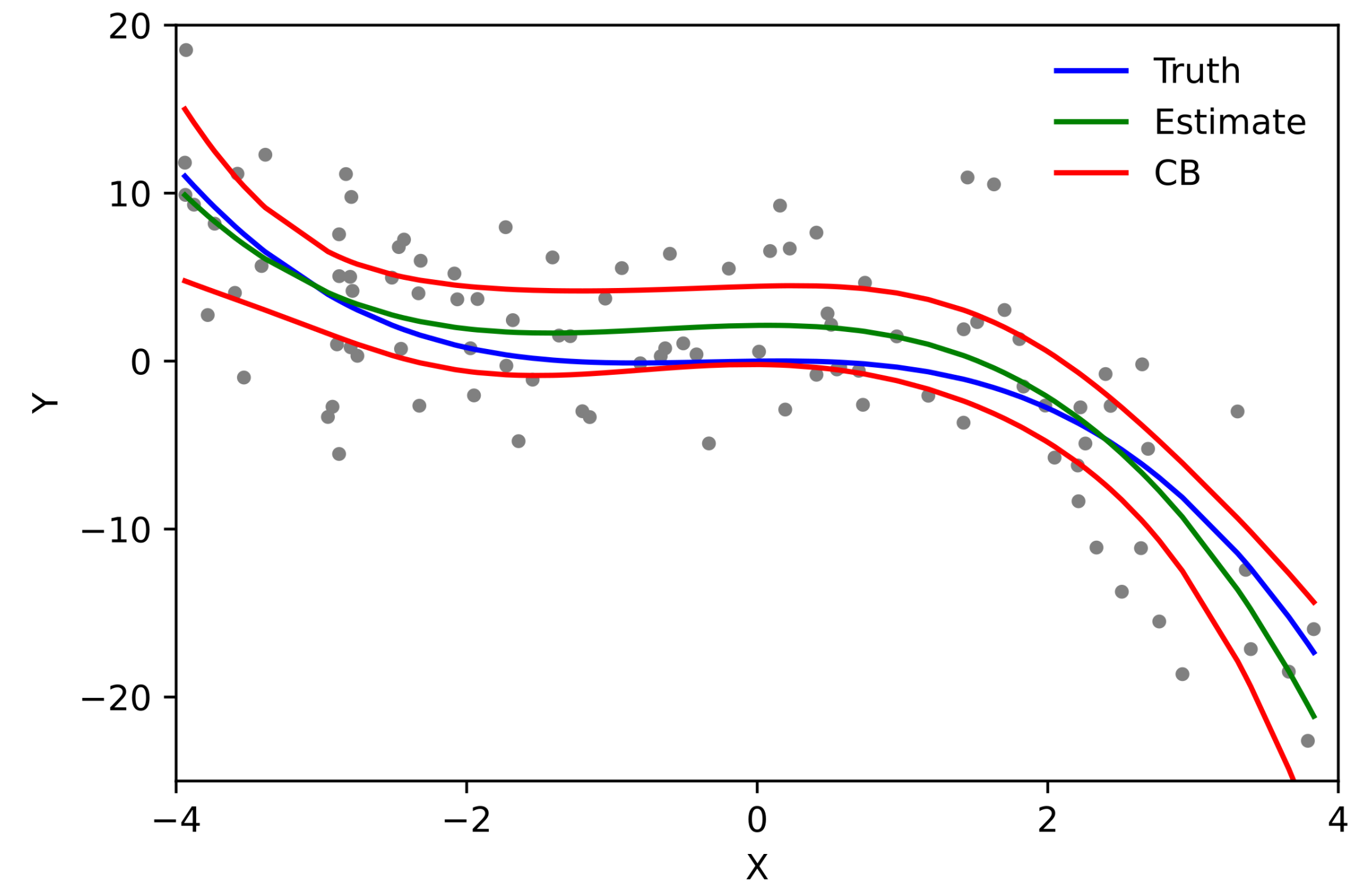
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What we are estimating is a *mathematical function*



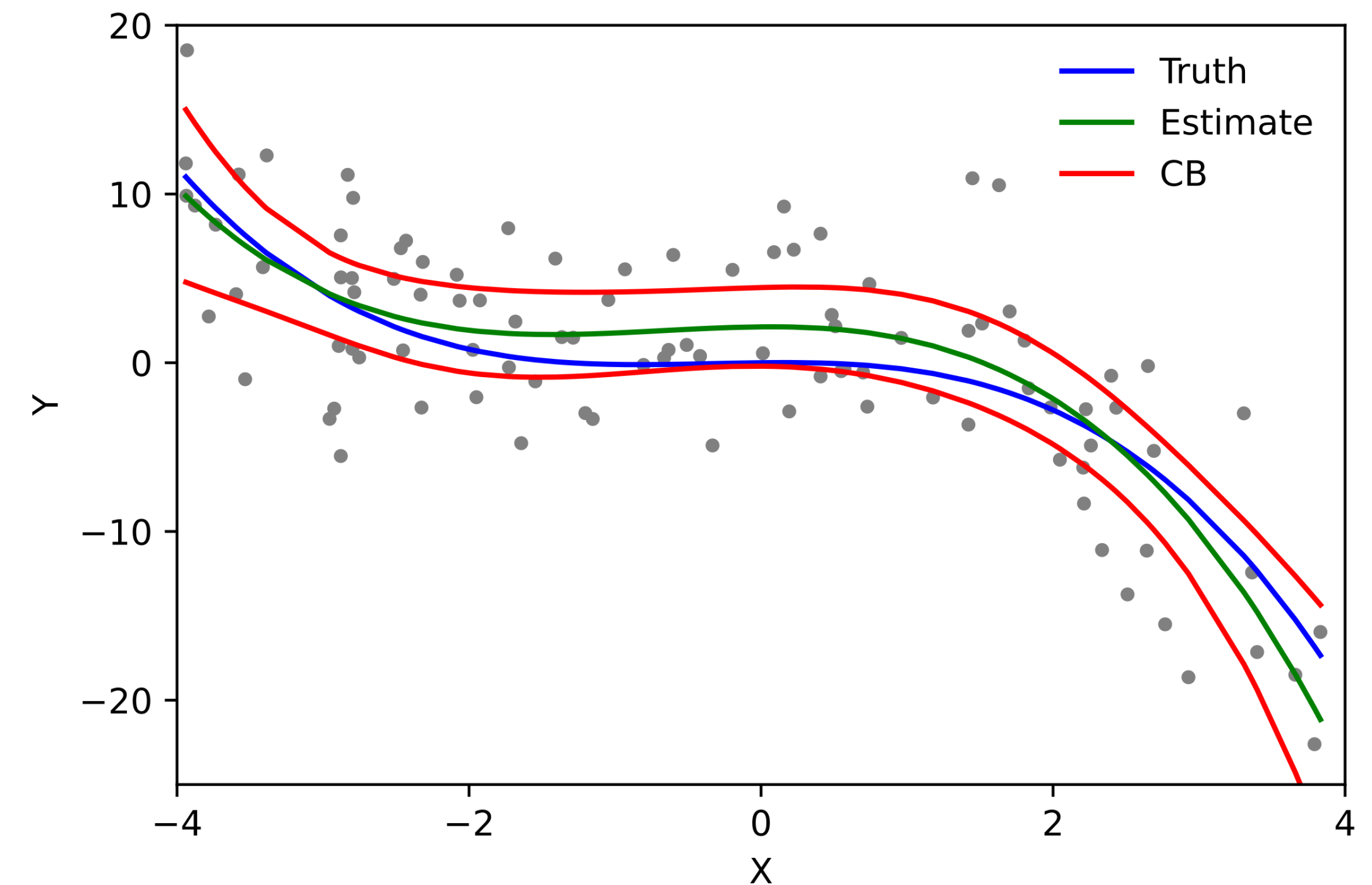


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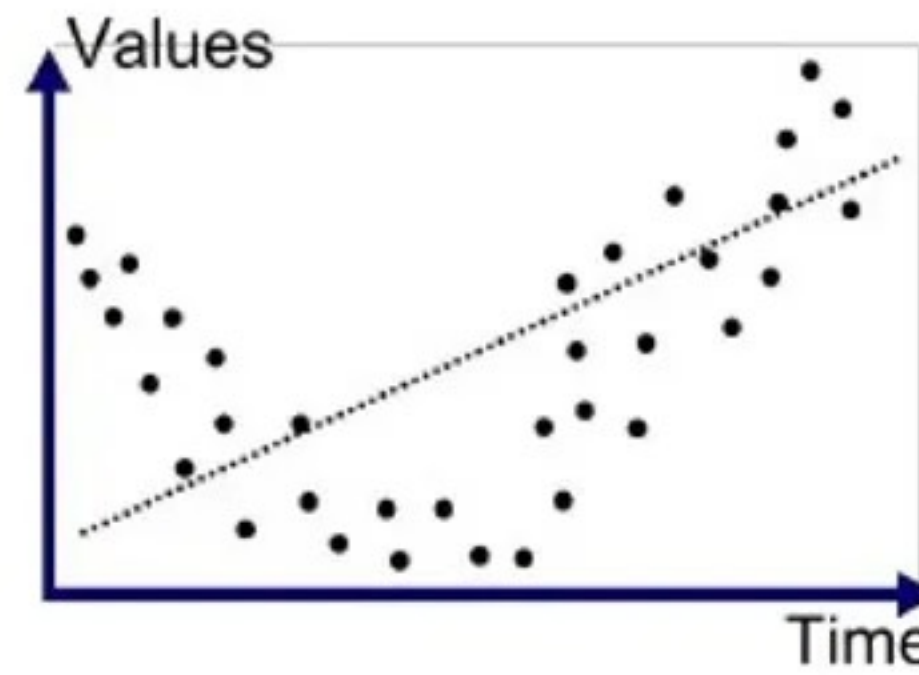
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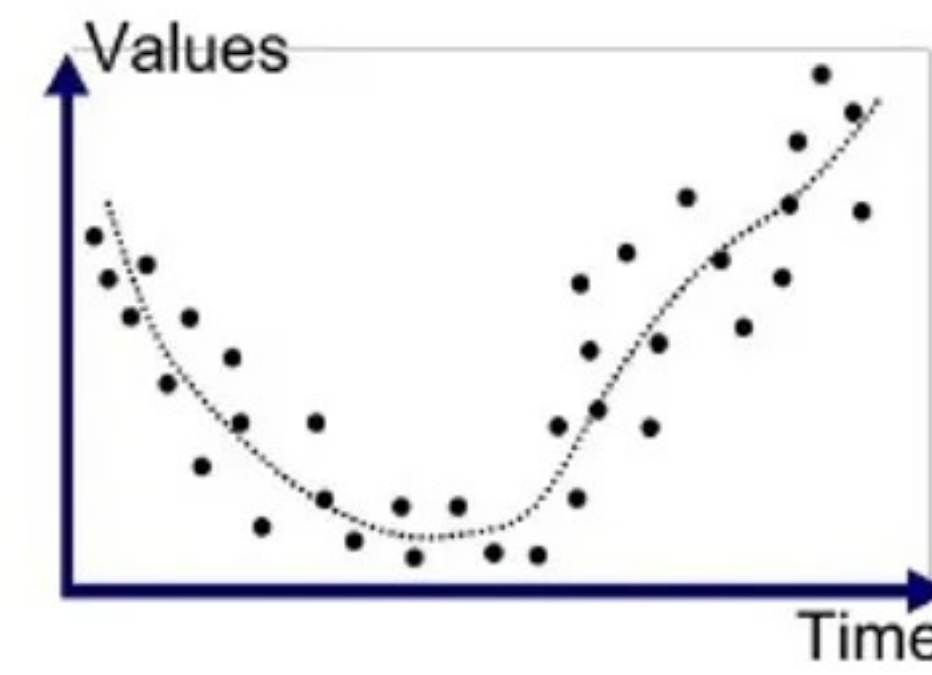
We think of the environment has providing us a function from our ind. variables to our dep. variables



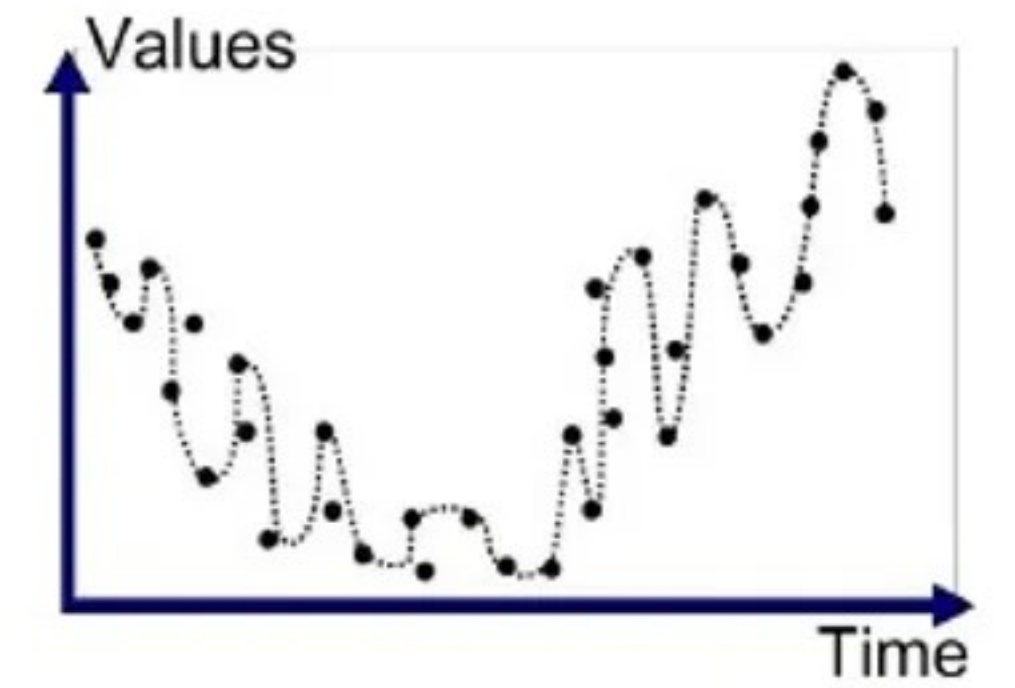
# Models



Underfitted

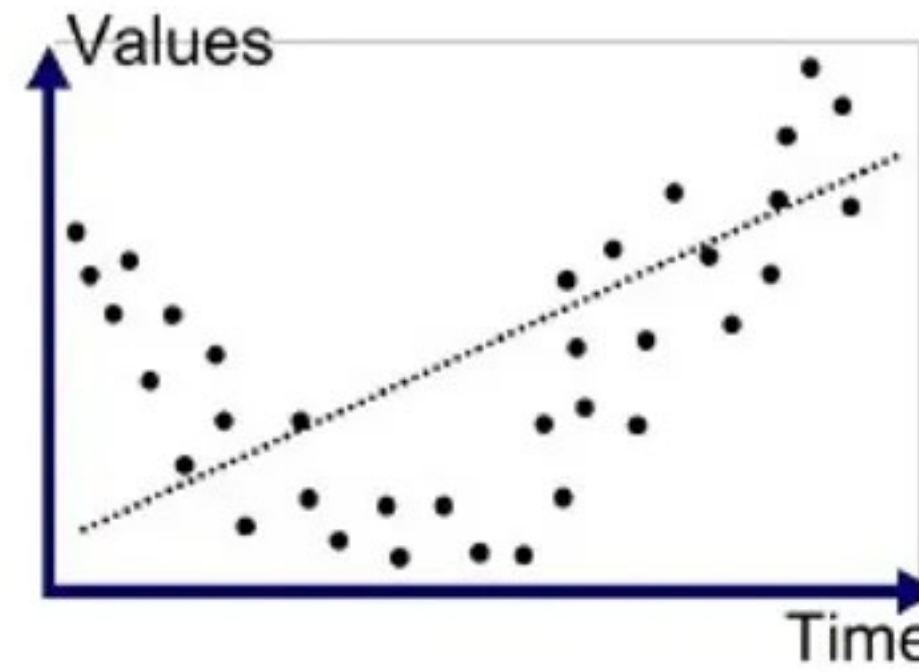


Good Fit/Robust

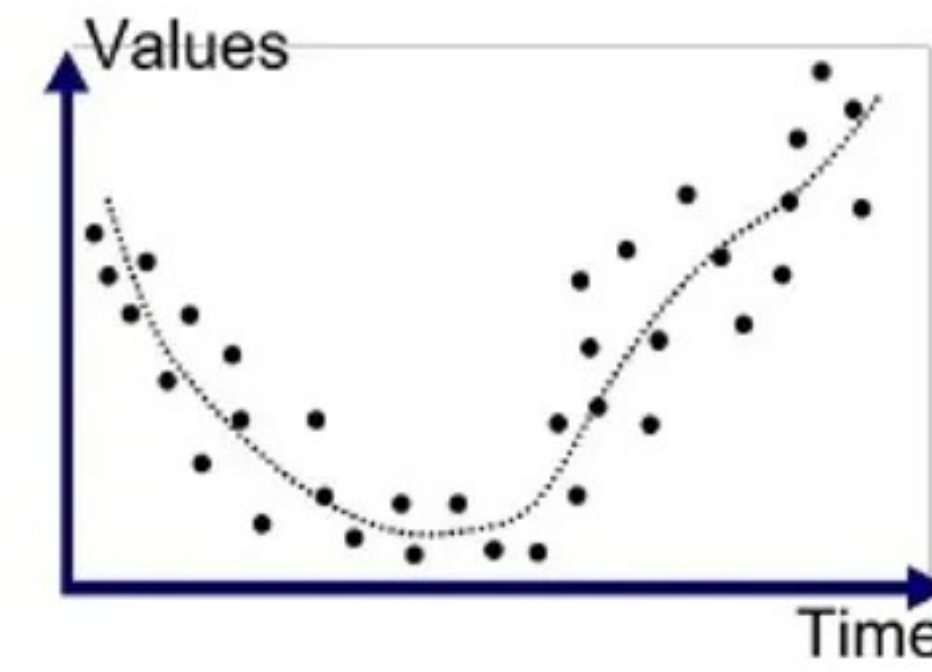


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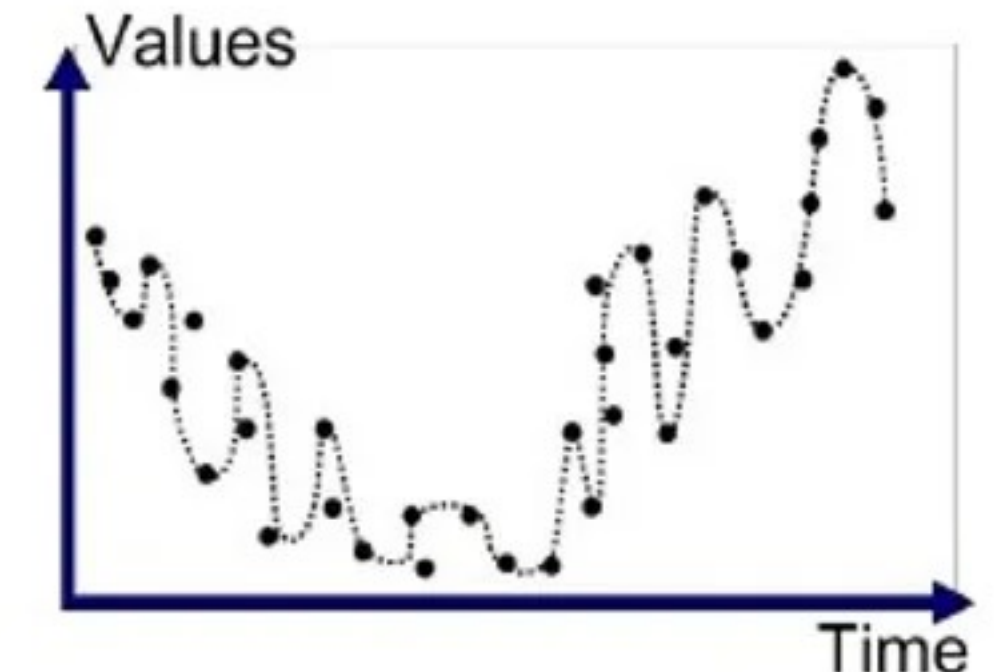
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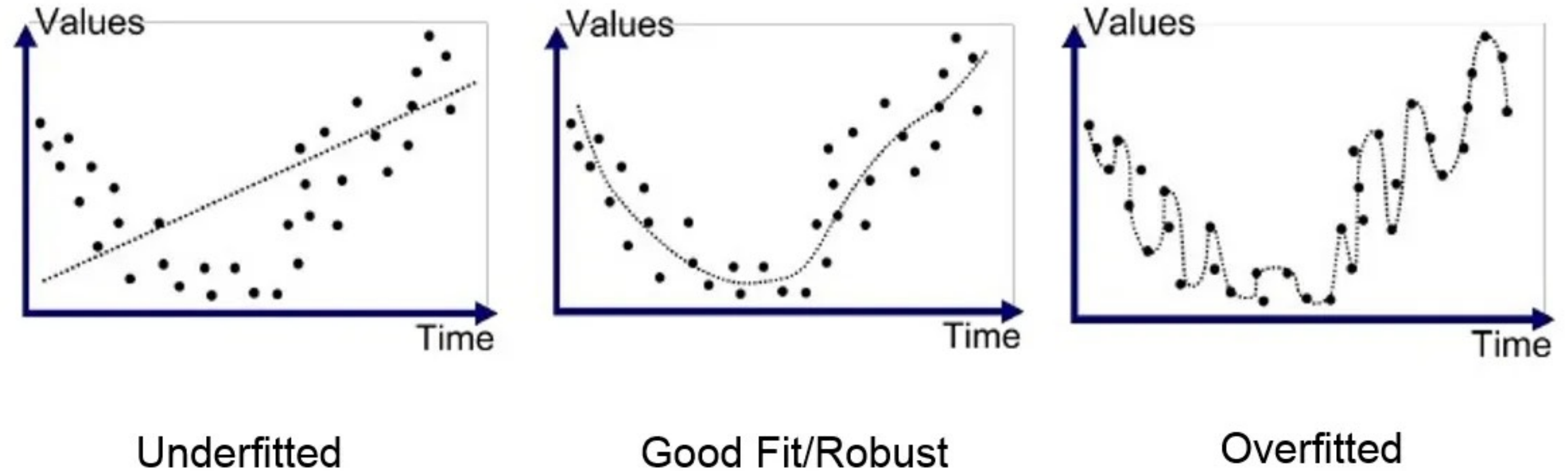


Overfitted

Therefore, a *model* is a mathematical function



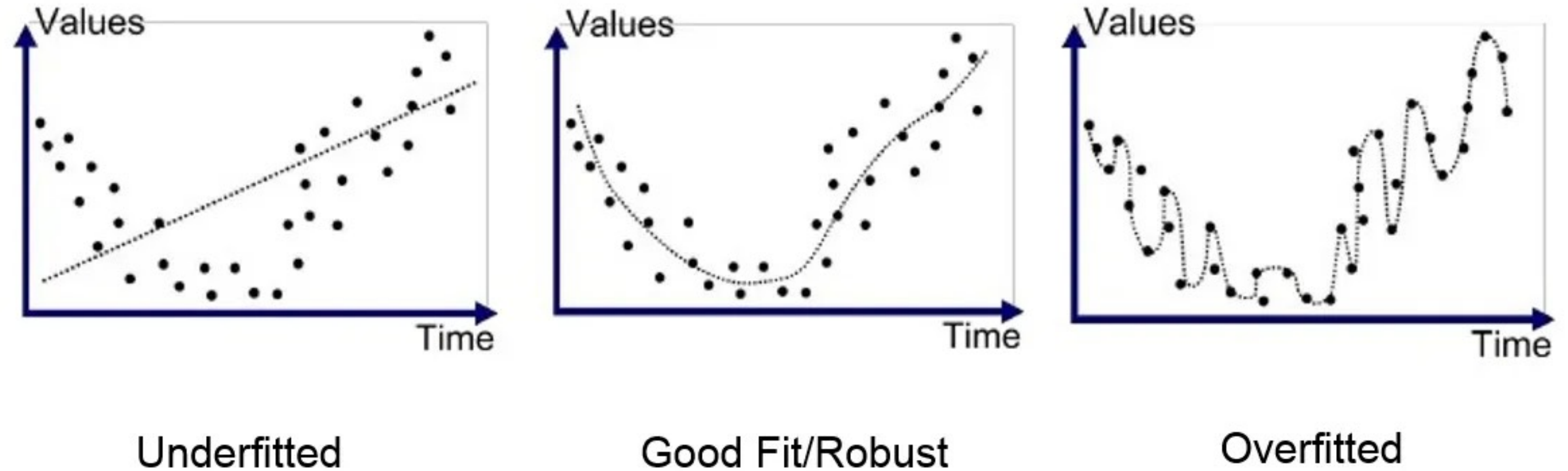
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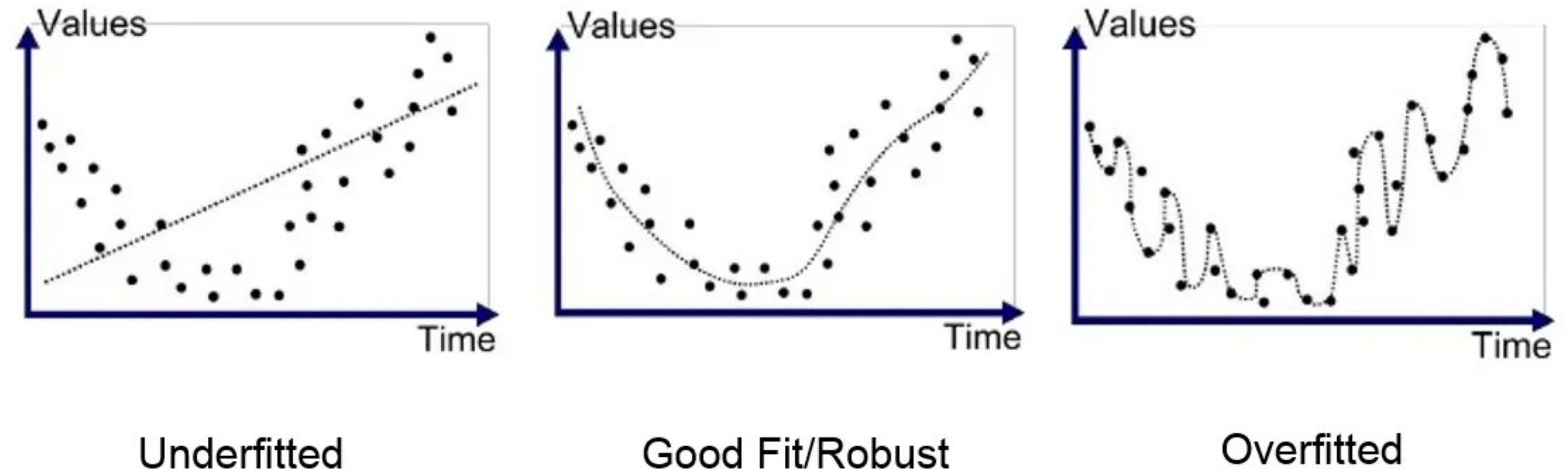
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(Advanced) We pick models from weaker classes of functions so that they are more robust when we **predict** values using the model

# How To: Prediction

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**Q.** Given  $\{(x_1, y_1), \dots, (x_k, y_k)\}$  use the line of best fit to predict the value of  $y'$  for the input  $x'$



# How To: Prediction

**Q.** Given  $\{(x_1, y_1), \dots, (x_k, y_k)\}$  use the line of best fit to predict the value of  $y'$  for the input  $x'$

**A.** Find the best fit line  $f(x) = \beta_0 + \beta_1 x$ . The predicted value of  $x'$  is  $f(x')$

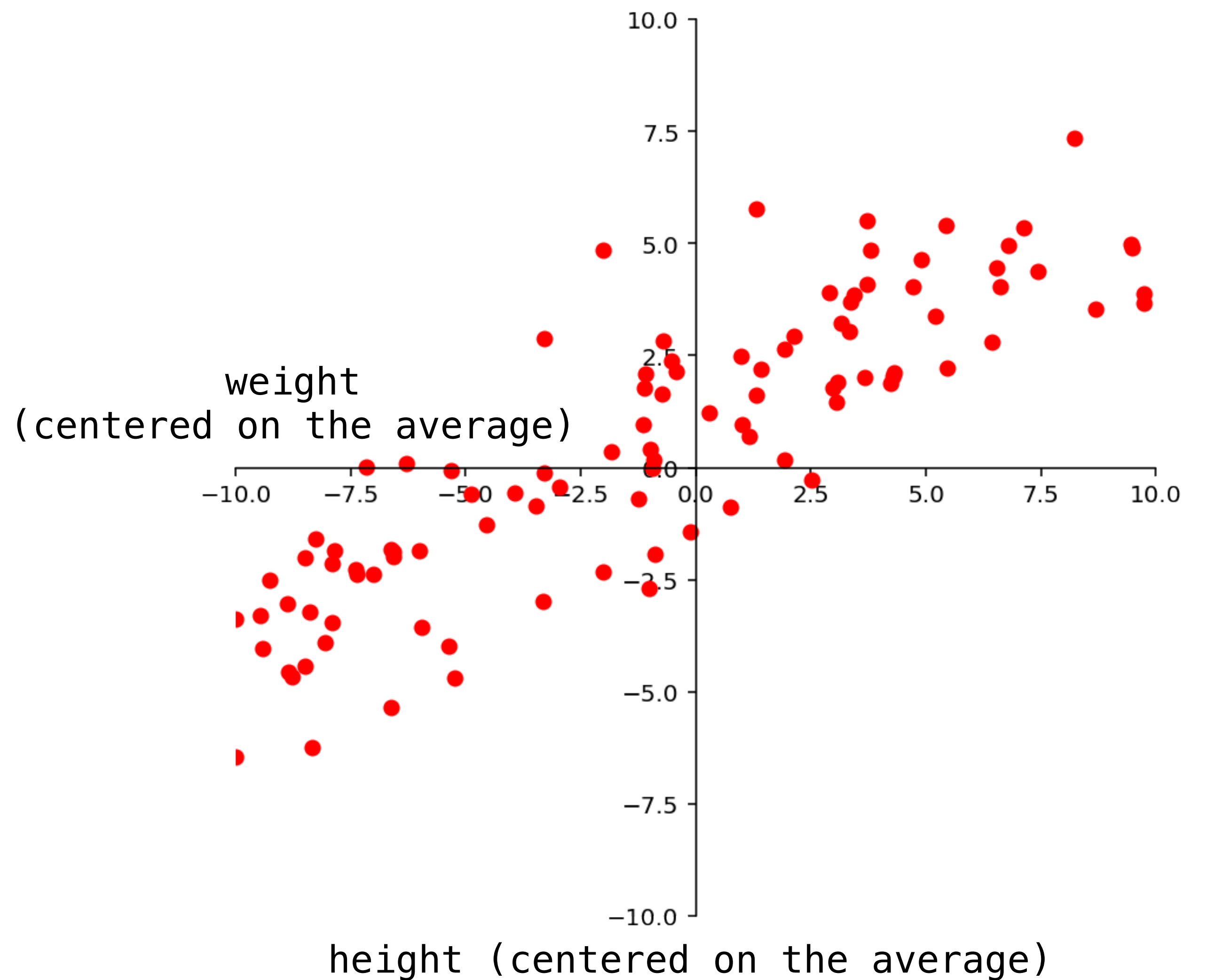
# How To: Prediction

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**This generalizes to any model fitting problem**

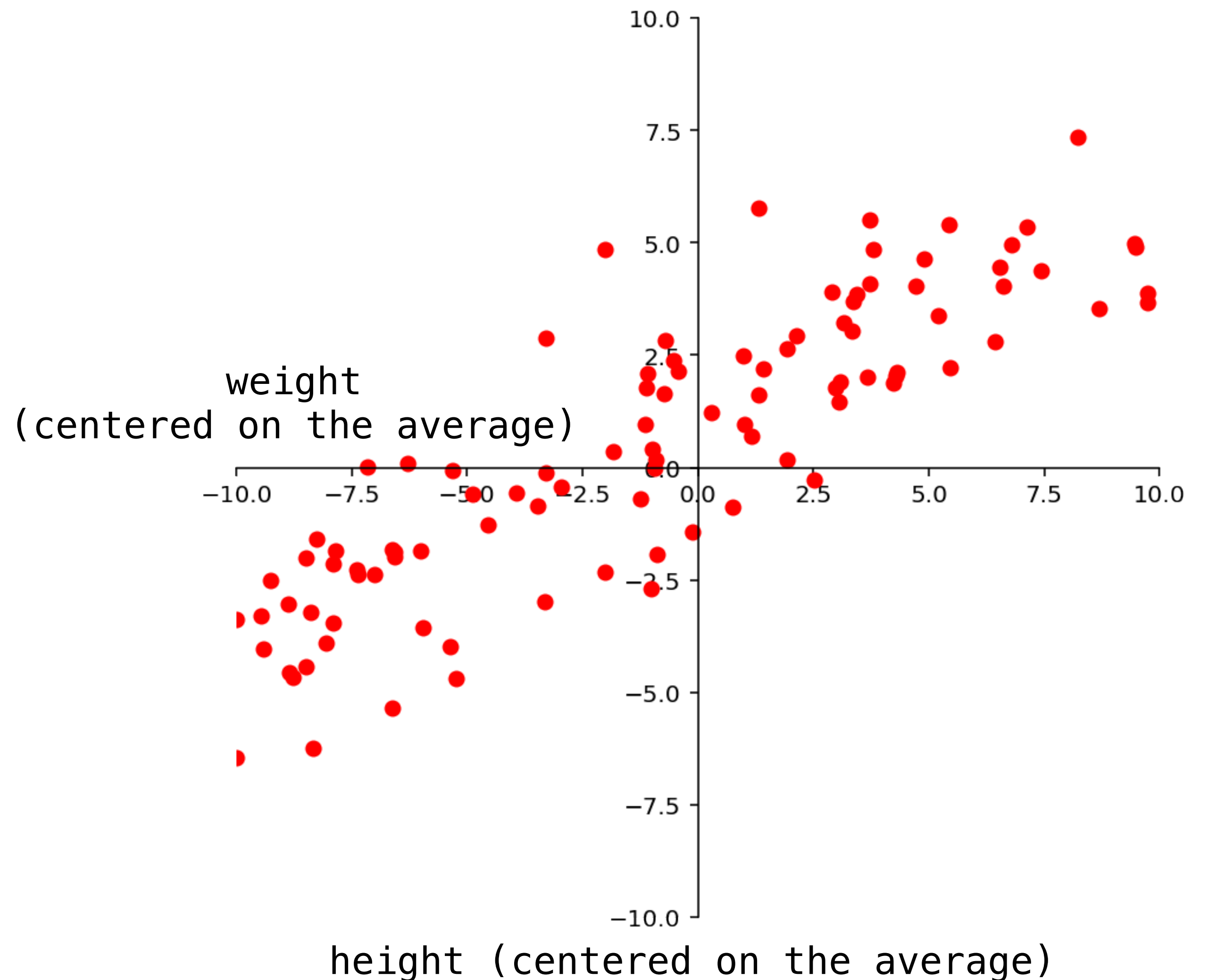
# Example: Height from Weight





# Example: Height from Weight

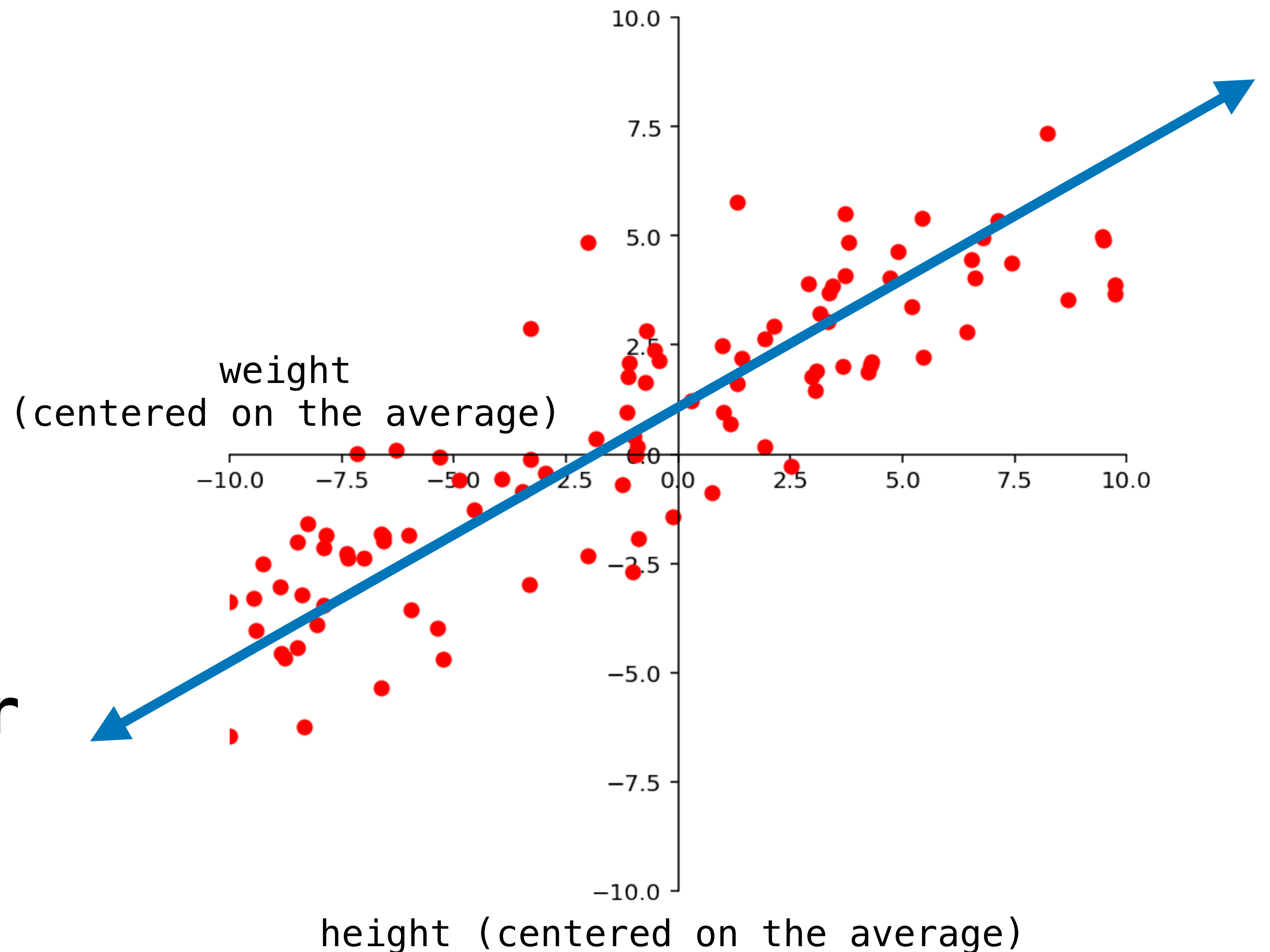
Suppose we know that person  $X$  weighs 150lb.  
*How would we guess the height of person  $X$ ?*



# Example: Height from Weight

Suppose we know that person  $X$  weighs 150lb. *How would we guess the height of person  $X$ ?*

If we know population data, then we can **find the line of best fit for that data** and then use that function

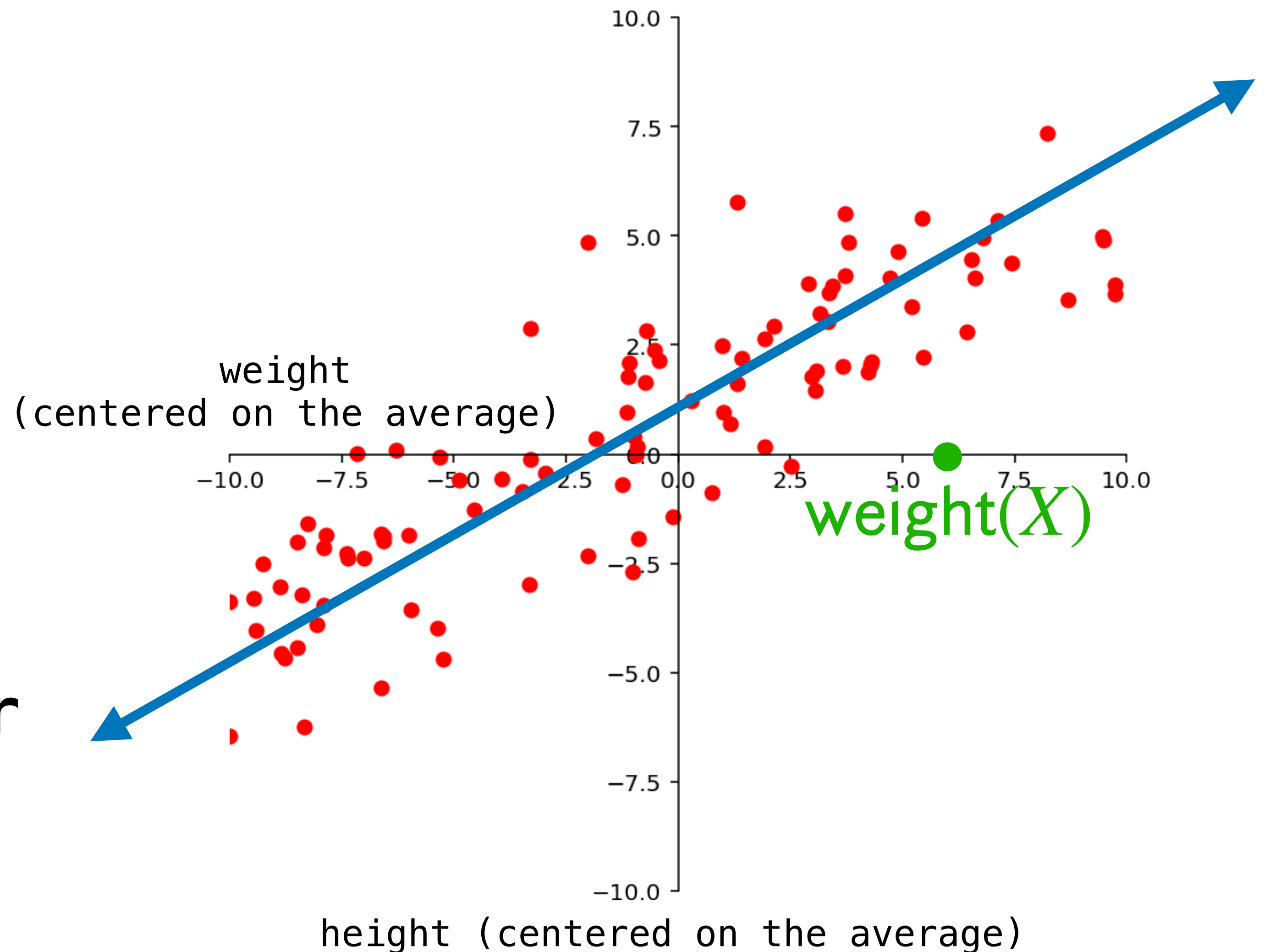




# Example: Height from Weight

Suppose we know that person  $X$  weighs 150lb. *How would we guess the height of person  $X$ ?*

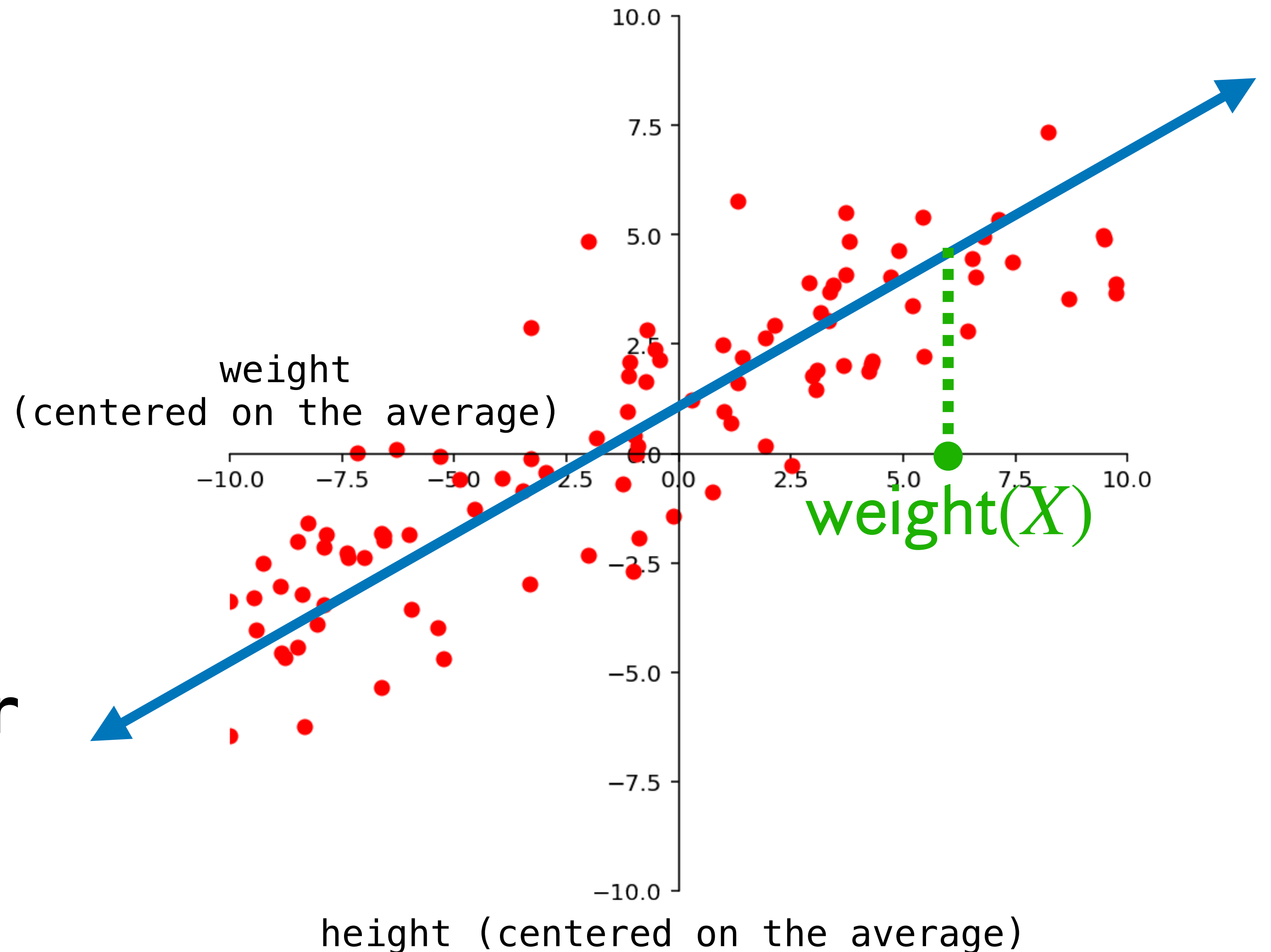
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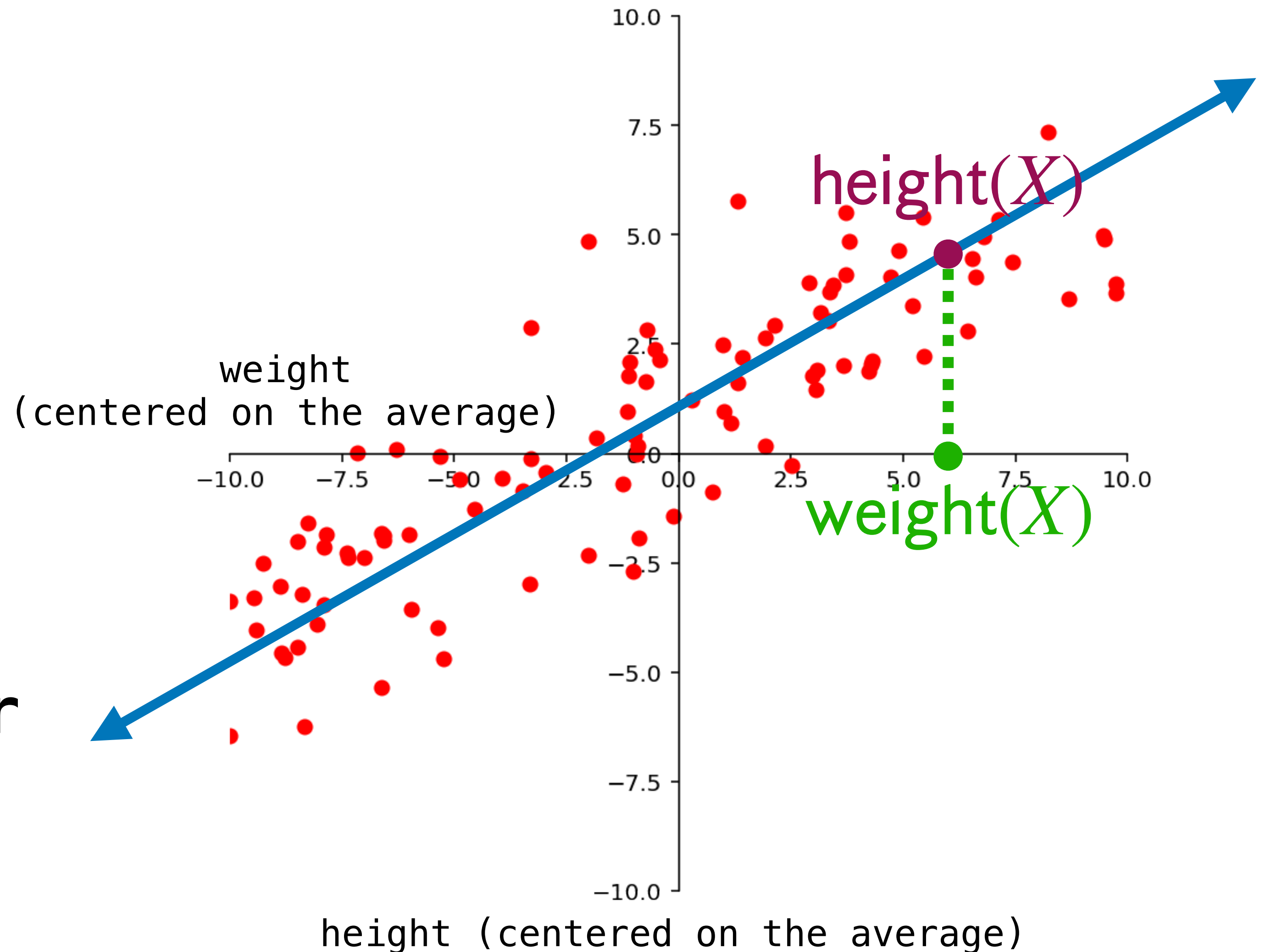
If we know population data, then we can **find the line of best fit for that data** and then use that function



# Example: Height from Weight

Suppose we know that person  $X$  weighs 150lb. *How would we guess the height of person  $X$ ?*

If we know population data, then we can **find the line of best fit for that data** and then use that function





# Question

*Find the line of best fit for the dataset*

$$\{(0,3), (1,1), (-1,1), (2,3)\}$$

*by using the the least-squares method*

*If you have time, graph your result and use it to "predict" the corresponding value for the input 4*

**Answer**

$$\{(0,3), (1,1), (-1,1), (2,3)\}$$

# **Linear Models and Least Squares Regression**



# **"Vectors" of Generalization**

# "Vectors" of Generalization

1. What if we have *more than one* independent value?



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**multiple regression, (hyper)plane of best fit**

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**e.g., polynomial regression**



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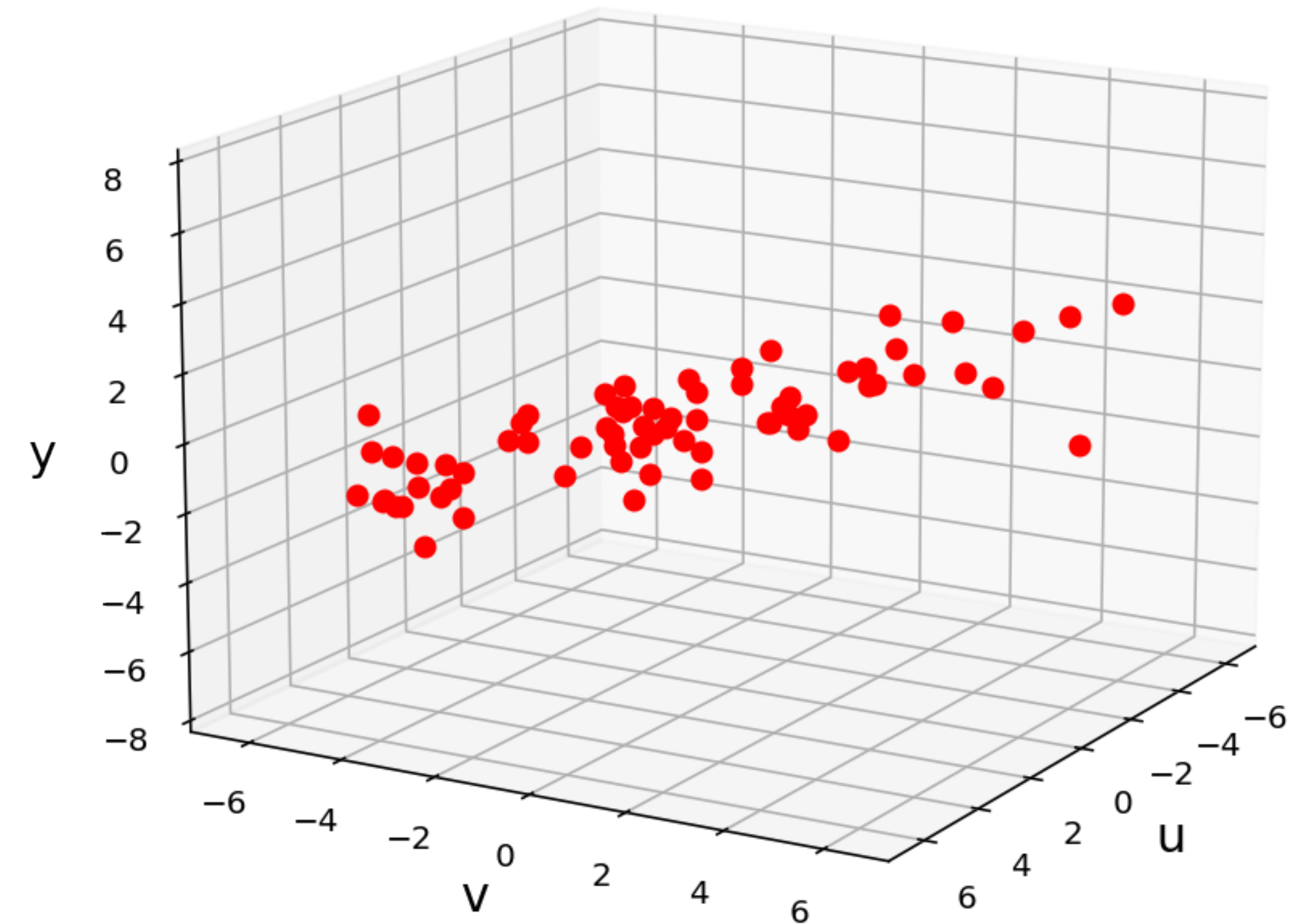
# Example: Terrain Data

Figure 23.1

Terrain Data for Multiple Regression

**Dataset:**  $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$   
where  $(x_i, y_i)$  is an longitude  
and latitude and  $z_i$  is an  
altitude

*Find the plane which  
"best" fits the data*



# Example: Terrain Data

Figure 23.2

Multiple Regression Fit to Data

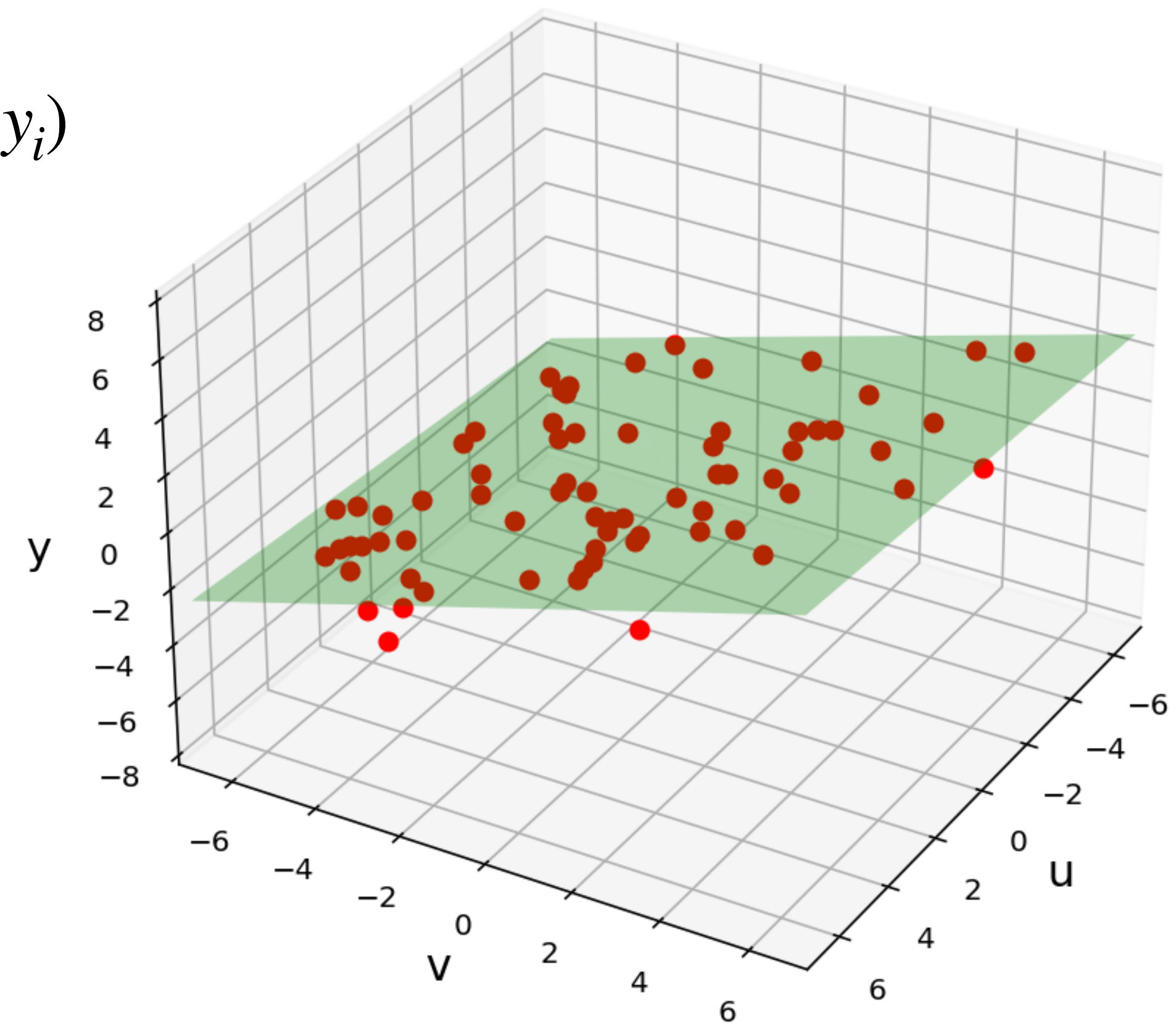
**Dataset:**  $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$  where  $(x_i, y_i)$  is an longitude and latitude and  $z_i$  is an altitude

Find  $\beta_0, \beta_1, \beta_2$  such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$





# Example: Terrain Data

Figure 23.2

Multiple Regression Fit to Data

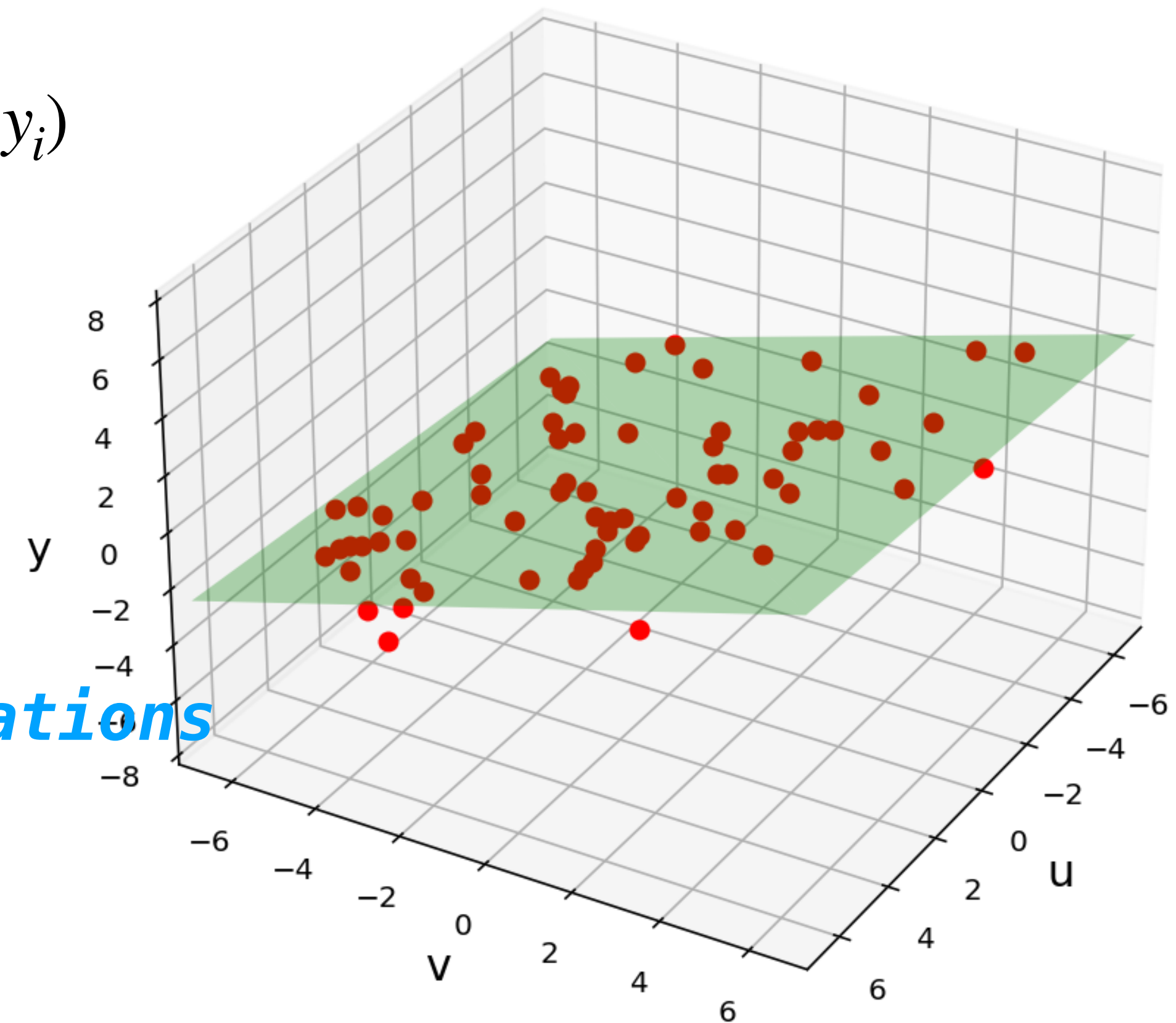
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*recall: planes are given by linear equations*  
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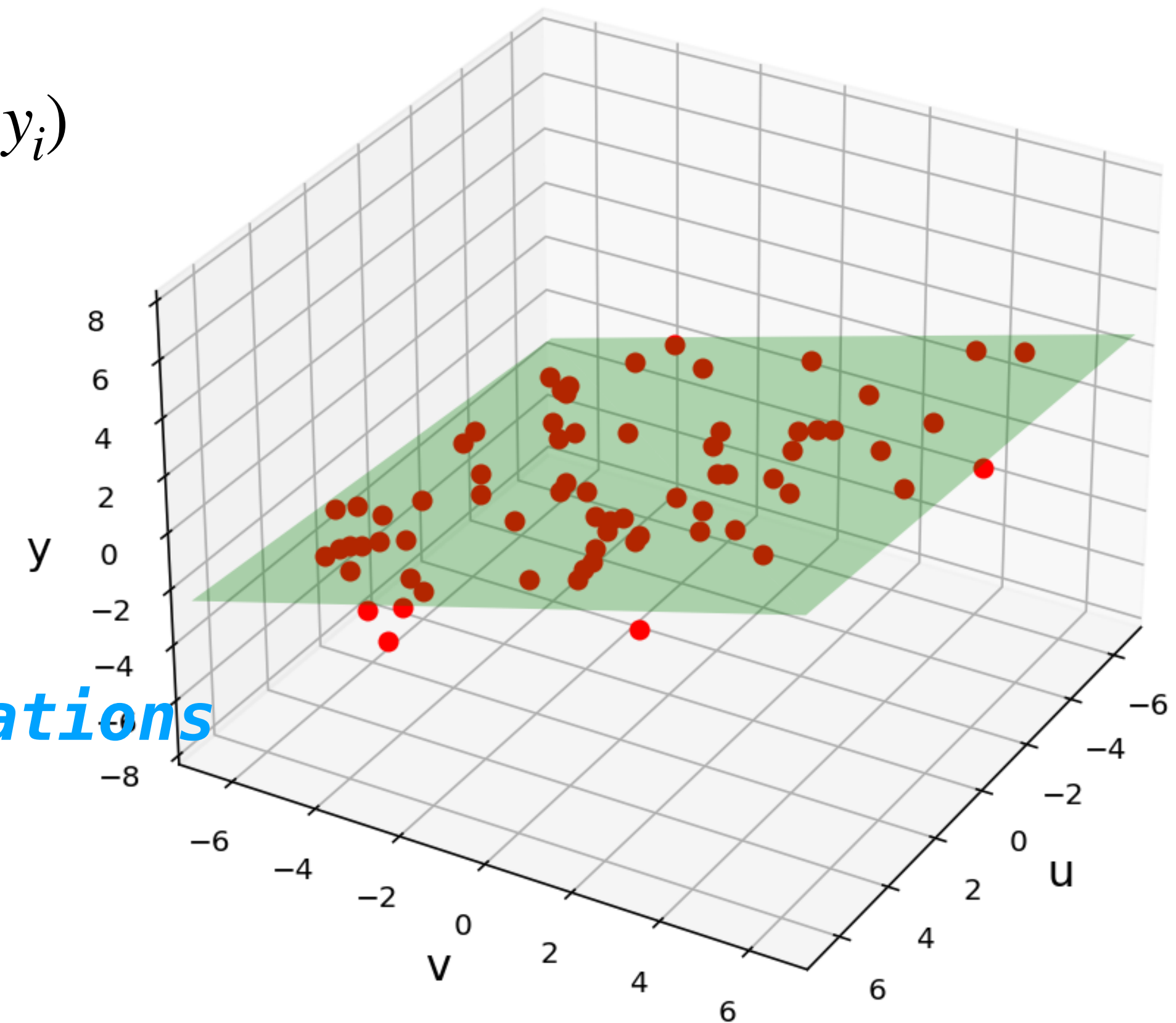
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*recall: planes are given by linear equations*  
which minimizes

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*$f(x, y)$  is a good approximation of the altitude*





# Example: Terrain Data

**Dataset:**  $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$  where  $(x_i, y_i)$  is an longitude and latitude and  $z_i$  is an altitude

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$$\beta_0 + \beta_1 x_1 + \beta_2 y_1 = z_1$$

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$\vdots$

$$\beta_0 + \beta_1 x_k + \beta_2 y_k = z_k$$

**Step 1:** Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables  $\beta_0, \beta_1, \beta_2$

# Example: Terrain Data

This is still linear in the  $\beta$ 's

**Dataset:**  $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$  where  $(x_i, y_i)$  is an longitude and latitude and  $z_i$  is an altitude

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$$\begin{bmatrix} 1 & \overset{\textcolor{blue}{X}}{x_1} & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_k & y_k \end{bmatrix} \begin{bmatrix} \overset{\textcolor{blue}{\vec{\beta}}}{\beta_0} \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \overset{\textcolor{blue}{Z}}{z_1} \\ z_2 \\ \vdots \\ z_k \end{bmatrix}$$

**Step 2:** Rewrite the system as a matrix equation



# Example: Terrain Data

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$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{z}$$

**Step 3:** Find the least squares solution of this system and use as the parameters of your model

# An Aside: Unique Least Squares

$$\begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_k & y_k \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{bmatrix}$$

***A (Conceptual).*** Why can we almost always assume that the columns of this matrix are LI?

# Answer



# Answer

If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

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First off, this is very unlikely



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Second, this variable could be then be thought of as a *dependent* variable

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If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

First off, this is very unlikely

Second, this variable could be then be thought of as a *dependent* variable

It wouldn't contribute anything when using the least squares method

# "Vectors" of Generalization



1. What if we have *more than one* independent value?

**multiple regression, (hyper)plane of best fit**

2. What if our data is not *exactly* linear.

**e.g., polynomial regression**



# "Vectors" of Generalization



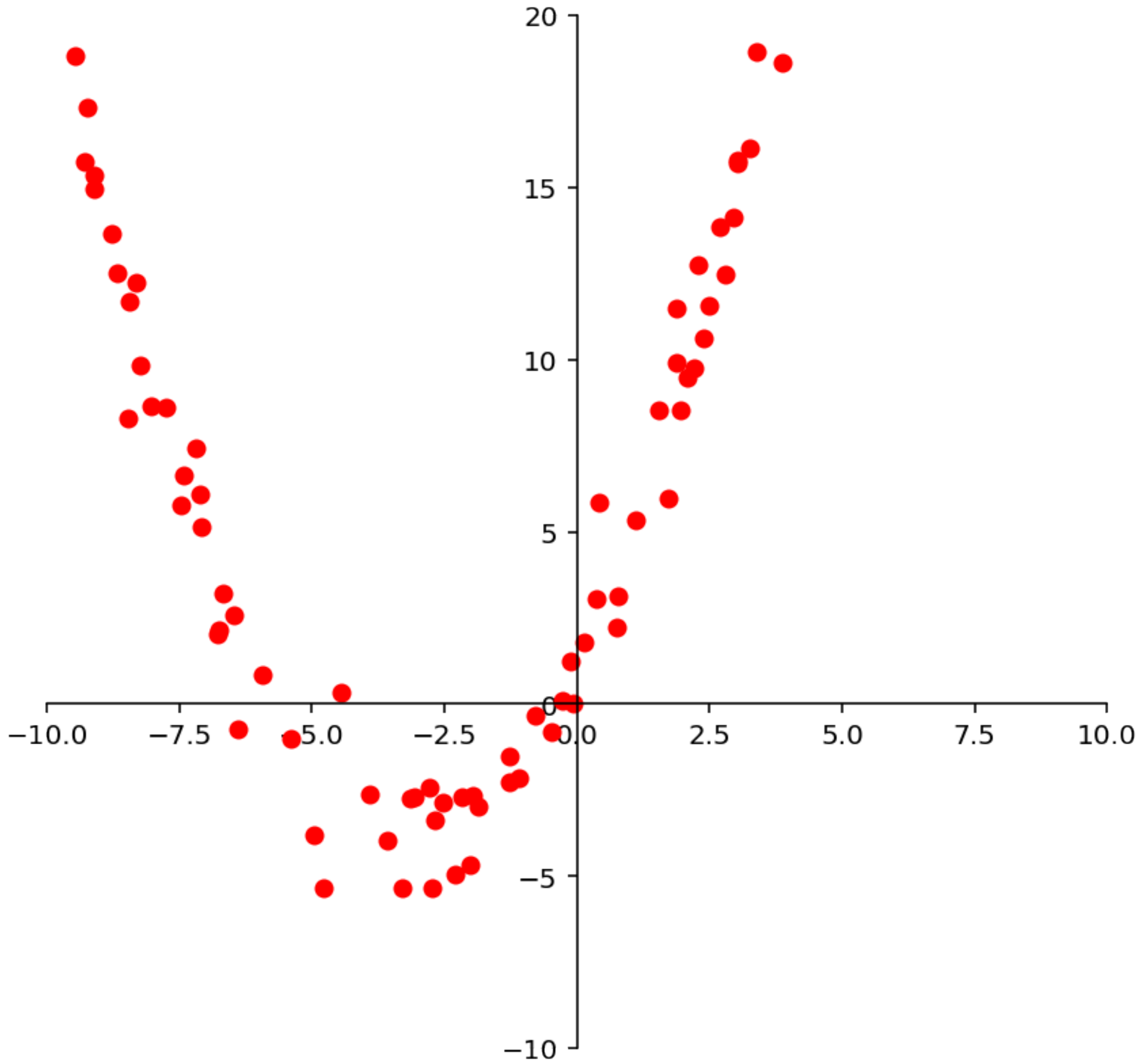
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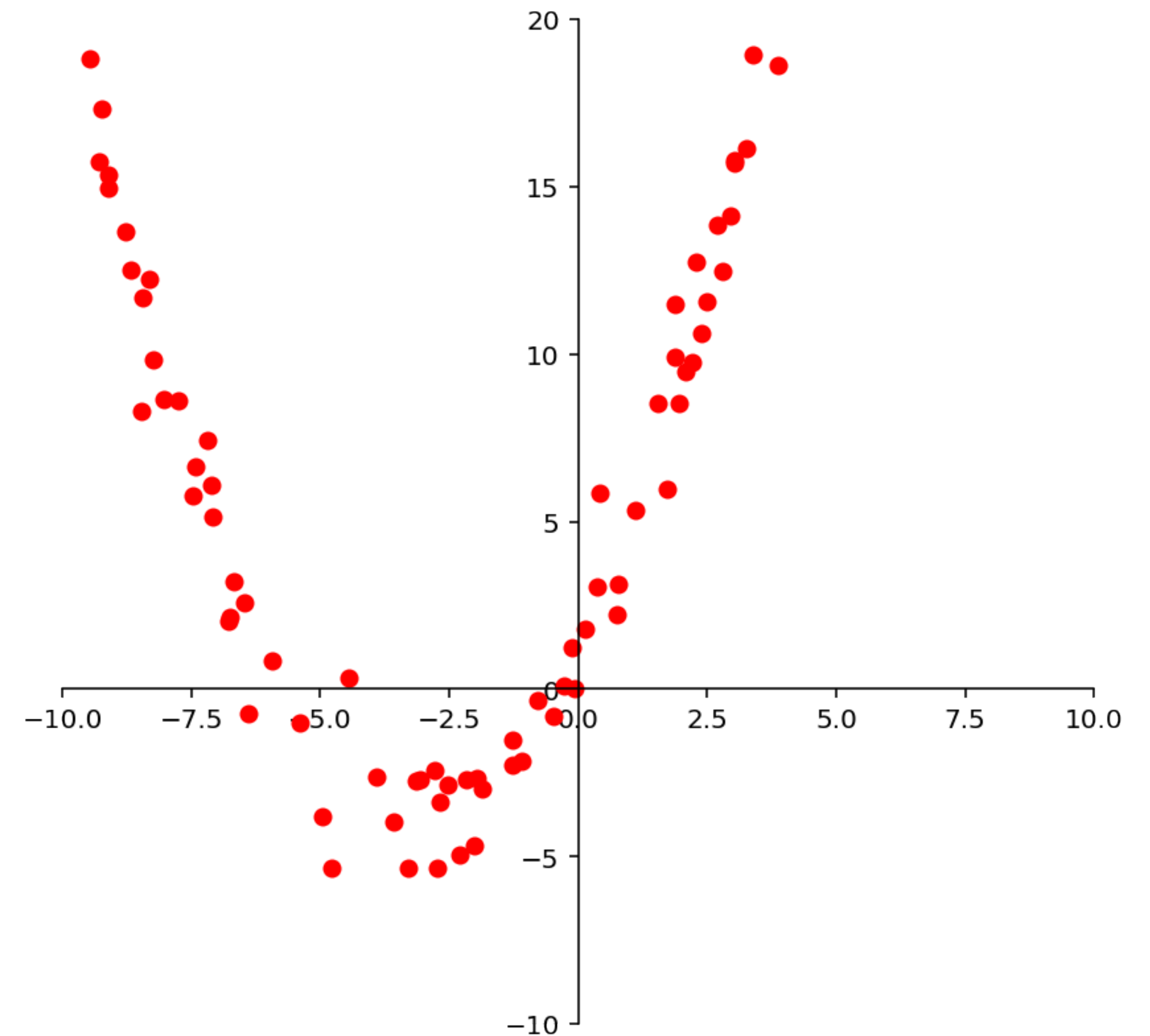
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# Example: Best Fit Quadratic



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**Dataset:**  $\{(x_1, y_1), \dots, (x_k, y_k)\}$

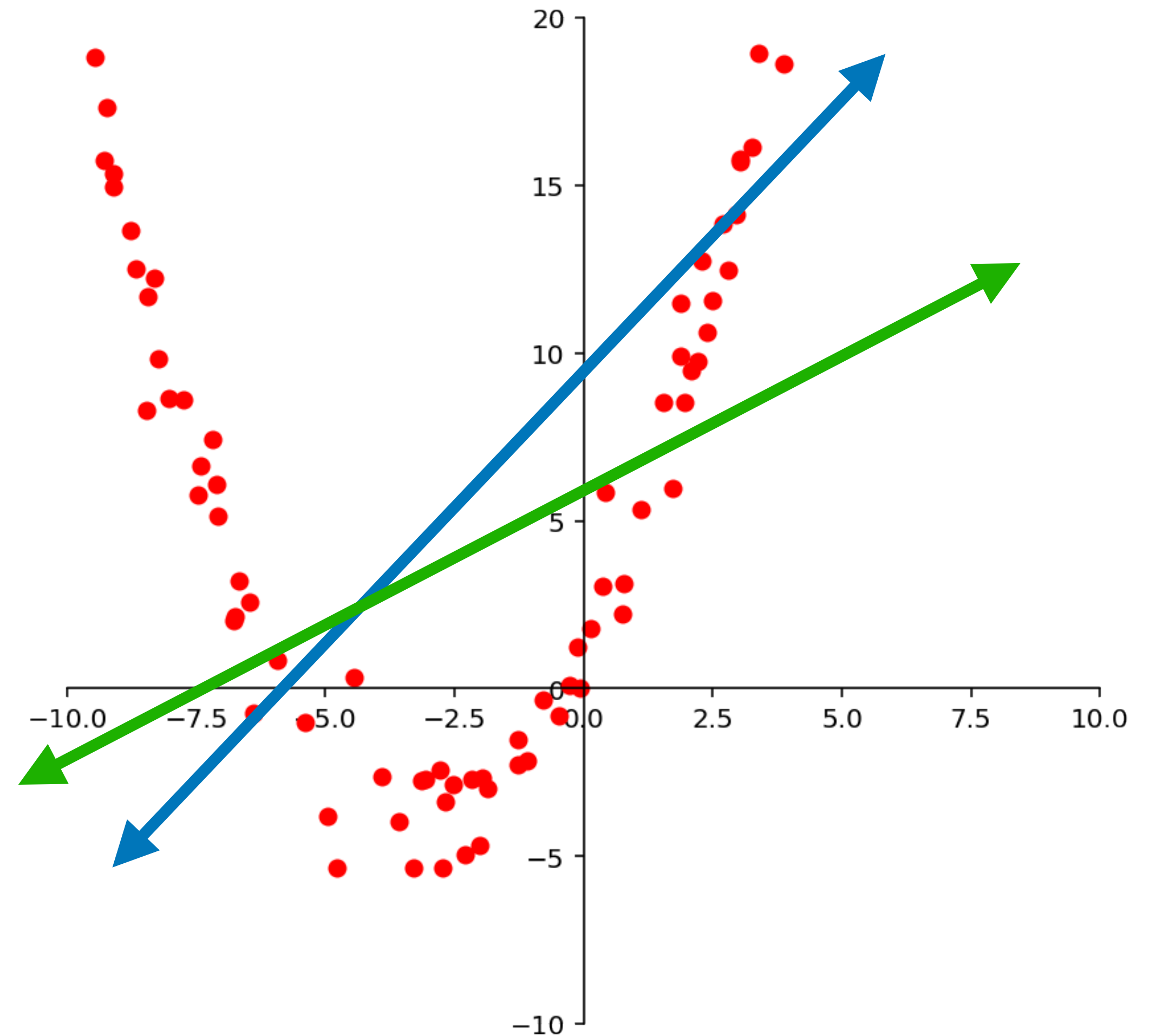




# Example: Best Fit Quadratic

**Dataset:**  $\{(x_1, y_1), \dots, (x_k, y_k)\}$

**The issue:** There is no good line to approximate this data

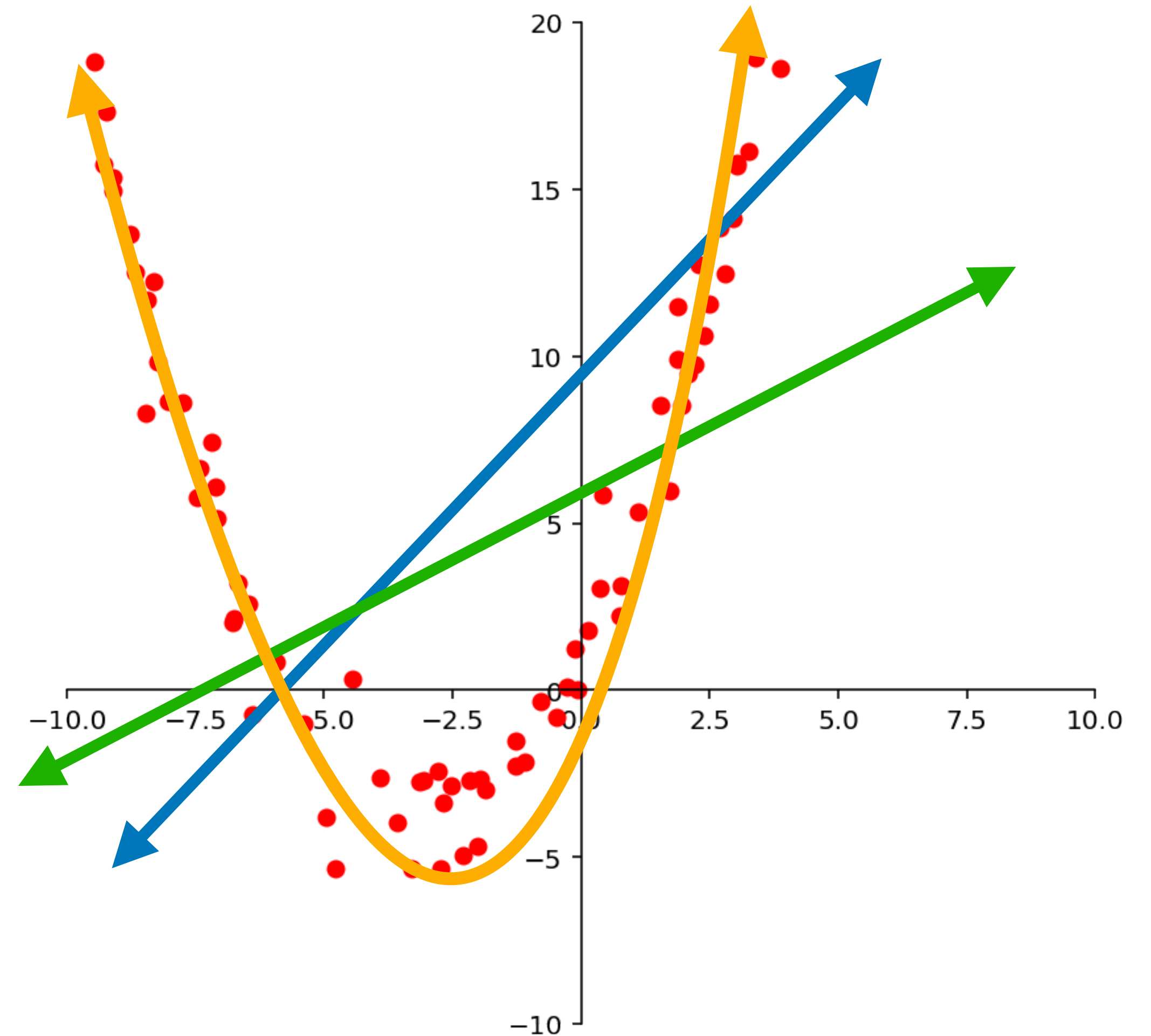


# Example: Best Fit Quadratic

**Dataset:**  $\{(x_1, y_1), \dots, (x_k, y_k)\}$

**The issue:** There is no good line to approximate this data

**What about a parabola?**





# Example: Best Fit Quadratic

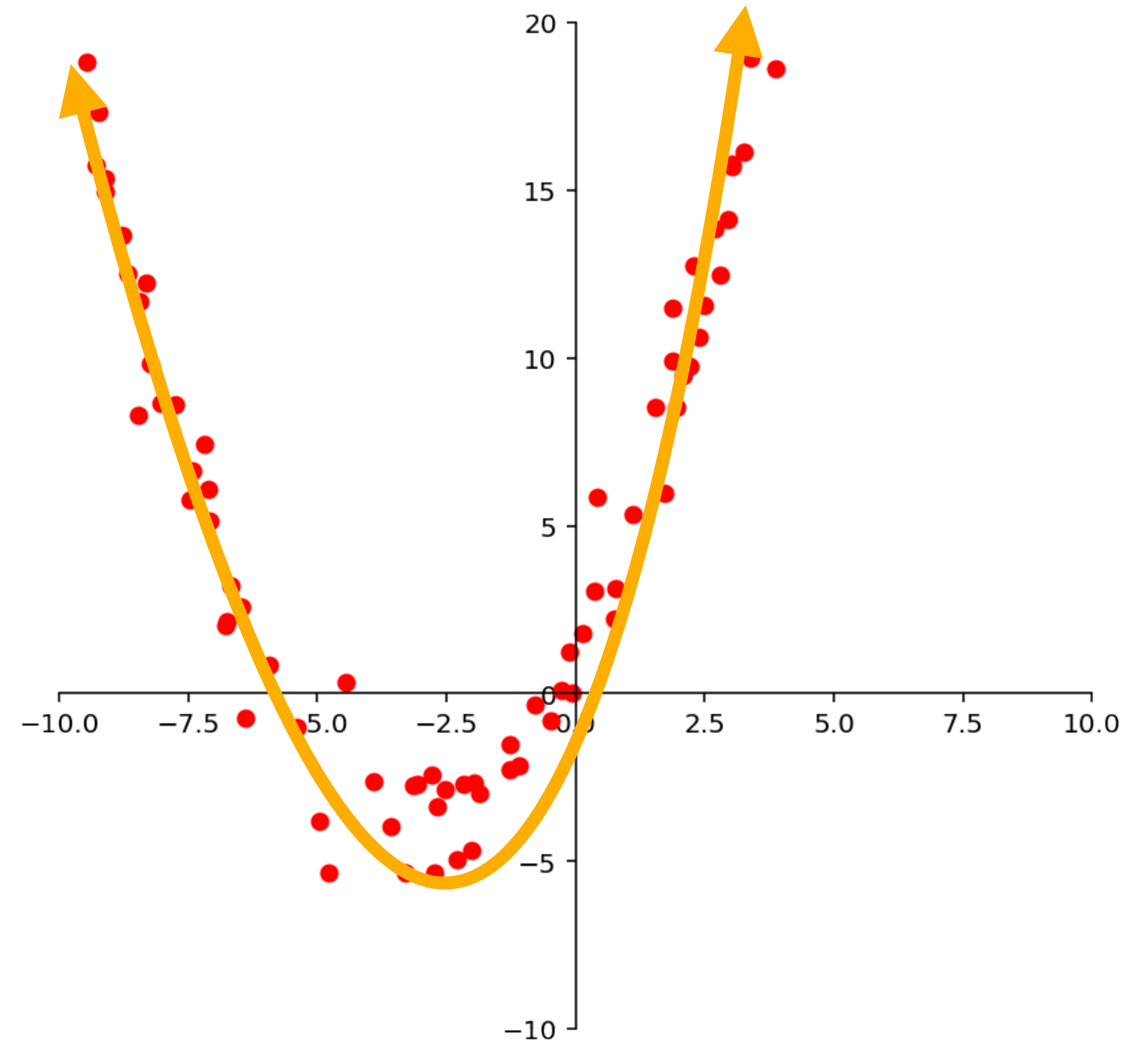
**Dataset:**  $\{(x_1, y_1), \dots, (x_k, y_k)\}$

Find  $\beta_0, \beta_1, \beta_2$  such that

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

minimizes

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$



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**Step 1:** Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables  $\beta_0, \beta_1, \beta_2$

# Example: Best Fit Quadratic

This is still linear in the  $\beta$ 's

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# Example: Best Fit Quadratic

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$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

**Step 3:** Find the least squares solution of this system and use as the parameters of your model

# The Takeaway

We can use non-linear modeling functions as long as they are *linear in the parameters*



# Linear in Parameters

**Def.** A function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is **linear in the parameters**  $\beta_1, \dots, \beta_k$  if it can be written as

$$f(\mathbf{x}) = \beta_1 \phi_1(\mathbf{x}) + \beta_2 \phi_2(\mathbf{x}) + \dots + \beta_k \phi_k(\mathbf{x})$$

for functions  $\phi_1, \dots, \phi_k: \mathbb{R}^n \rightarrow \mathbb{R}$

We can build design matrices for  
function which are linear in  
their parameters



# General Linear Regression

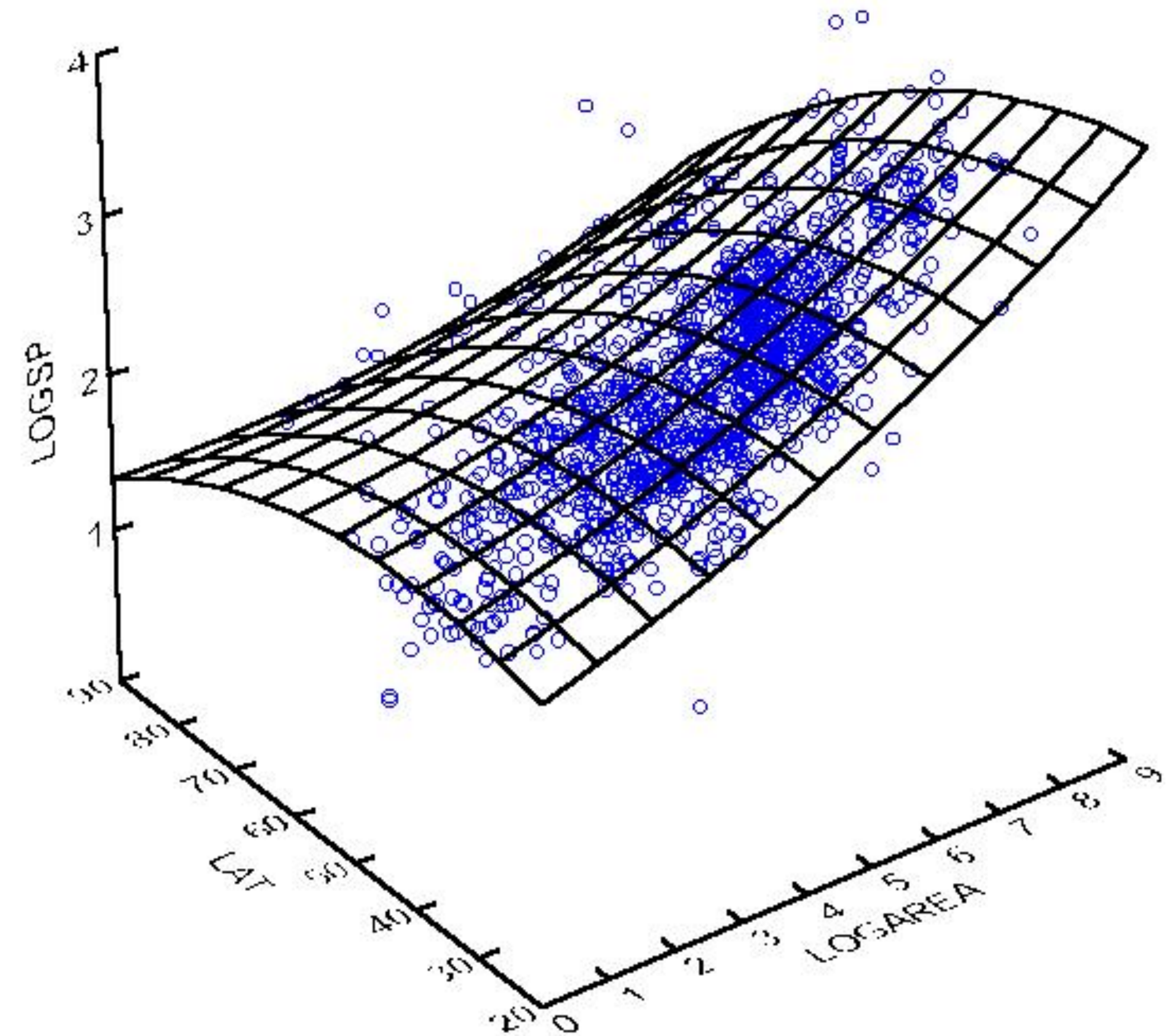
**Dataset:**  $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$  where  $\mathbf{x}_i \in \mathbb{R}^n$  and  $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters*  $\beta_1, \dots, \beta_k$ , find values for  $\beta_1, \dots, \beta_k$  which minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$



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$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$

$$\beta_1 \phi_1(\mathbf{x}_1) + \dots + \beta_k \phi_k(\mathbf{x}_1) = y_1$$

$$\beta_1 \phi_1(\mathbf{x}_2) + \dots + \beta_k \phi_k(\mathbf{x}_2) = y_2$$

$\vdots$

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**Step 1:** Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables  $\beta_1, \dots, \beta_k$

# General Linear Regression

This is still linear in the  $\beta$ 's

**Dataset:**  $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$  where  $\mathbf{x}_i \in \mathbb{R}^n$  and  $y_i \in \mathbb{R}$

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$$\begin{aligned} \beta_1 \phi_1(\mathbf{x}_1) + \dots + \beta_k \phi_k(\mathbf{x}_1) &= y_1 \\ \beta_1 \phi_1(\mathbf{x}_2) + \dots + \beta_k \phi_k(\mathbf{x}_2) &= y_2 \\ &\vdots \\ \beta_1 \phi_1(\mathbf{x}_m) + \dots + \beta_k \phi_k(\mathbf{x}_m) &= y_m \end{aligned}$$

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# General Linear Regression

**Dataset:**  $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$  where  $\mathbf{x}_i \in \mathbb{R}^n$   
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design matrix

$$\overset{X}{\begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \dots & \phi_k(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \dots & \phi_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & \dots & \phi_k(\mathbf{x}_m) \end{bmatrix}} \begin{bmatrix} \vec{\beta} \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} = \begin{bmatrix} \mathbf{y} \\ y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}$$

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# General Linear Regression

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**Step 3:** Find the least squares  
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# How To: Design Matrices

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**Q.** Find the design matrix for least squares regression with the function  $f$  in terms of the parameters  $\beta_1, \beta_2, \dots, \beta_k$  given the dataset  $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$



# How To: Design Matrices

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**A.** First write  $f(\mathbf{x})$  as  $\beta_1\phi_1(\mathbf{x}) + \dots + \beta_k\phi_k(\mathbf{x})$  where  $\phi_1, \dots, \phi_k$  are potentially non-linear functions. Then build the matrix:

$$\begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \dots & \phi_k(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \dots & \phi_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & \dots & \phi_k(\mathbf{x}_m) \end{bmatrix}$$



# Example

*Find the design matrix for the least squares regression with the function*

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \beta_1 \cos(x_1) + \beta_2 e^{-x_1 x_2} - \beta_1 x_3 + \beta_3$$

*for the dataset*

$$\mathbf{x}_1 = (0,0,0) \quad y_1 = 5$$

$$\mathbf{x}_2 = (\pi,3,1) \quad y_2 = 3$$

We will see more of this  
in Lab 6

# Practical Considerations



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Many functions require large design matrices, e.g.  
multivariate polynomials have *a lot* of possible terms

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Many functions require large design matrices, e.g.  
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We haven't actually talked about *which* modeling  
functions to use

# Practical Considerations

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Again, is least-squares error really what we want? What if we want to minimize something else?



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**Concerns for another class**